

Analysis and Control of Hindmarsh-Rose Neuronal model

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Chapter 1

Introduction

The regulation of neuronal electrical activity is a critical objective for addressing pathologies such as Parkinson’s disease and epilepsy. Effective neurostimulation necessitates mathematical models that balance analytical tractability with biological complexity [1]. The Hindmarsh-Rose (HR) model fulfills these requirements, accurately capturing a diverse range of behaviors including tonic spiking and chaotic bursting [3].

Initial investigation focuses on the derivation of the HR model from the Hodgkin-Huxley equations and its characteristic slow-fast decomposition [1]. A systematic stability analysis is conducted using numerical analysis and Routh-Hurwitz conditions, alongside numerical bifurcation mapping to identify key transitions. These include subcritical Hopf bifurcations and chaotic windows, which are characterized via Inter-Spike Interval (ISI) diagrams and Poincaré sections following established methodologies [2].

Control synthesis is achieved by incorporating an external current into the membrane potential equation to simulate clinical interventions like Deep Brain Stimulation. Three primary strategies are evaluated: a linearized PID controller, exact Input-Output Feedback Linearization, and robust Sliding Mode Control. Beyond setpoint regulation, these methods are applied to the tracking/synchronization of a leader-follower pair. This is addressed through a pseudo pinning control framework, where the controller forces a follower neuron to track the firing trajectory dictated by a leader neuron.

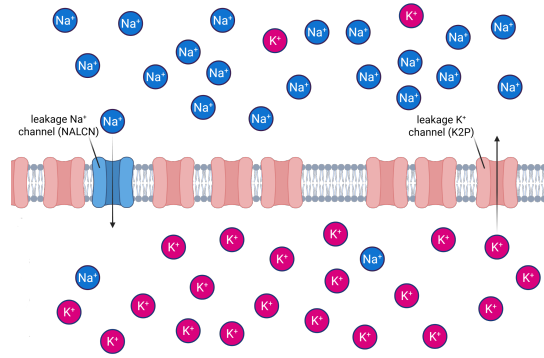
Finally, controller performance is assessed under conditions of parametric uncertainty and input disturbances. This study concludes by suggesting future research directions.

Disclaimer: This report was entirely conceived, structured, and authored by the author. Assistance from the Gemini AI model was strictly limited to linguistic refinement and enhancing the overall clarity of exposition. All mathematical models, numerical simulations, and analytical results were developed independently by the author without reliance on AI-generated content. The supporting software—which incorporates snippets and methodologies derived from previous experience in university associations and roles within the robotics world—is provided under the GPL license. All simulations were performed using Matlab 2024b on Ubuntu 24.04. Furthermore, nearly all figures are provided in a lossless vector format, allowing for high-level magnification without loss of image quality.

Chapter 2

Context

The Hindmarsh-Rose (HR) model [3] is a simplified mathematical and phenomenological model of neuronal activity consisting of three coupled first-order differential equations. It is widely used because of its low dimensionality, analytical tractability, and its ability to qualitatively capture a rich repertoire of complex neuronal behaviors, including quiescence, continuous tonic spiking, periodic bursting, and chaotic firing.



The model is defined by three coupled ordinary differential equations, commonly written as:

$$\begin{aligned}\dot{x}_1 &= x_2 - ax_1^3 + bx_1^2 - x_3 + I \\ \dot{x}_2 &= c - dx_1^2 - x_2 \\ \dot{x}_3 &= r[s(x_1 - x_r) - x_3]\end{aligned}\tag{2.1}$$

where, the variables (x_1, x_2, x_3) :

- x_1 (**Membrane Potential**): This represents the voltage across the neuronal membrane.
- x_2 (**Recovery Variable**): Represents the fast-acting ion currents (such as the potassium K^+ channels) that pull the voltage back down after a spike.
- x_3 (**Adaptation Current**): This is the slow variable. It changes much more slowly than x_1 or x_2 (controlled by the small parameter r). It is responsible for the "bursting" behavior by slowly increasing during a burst until it inhibits the firing, causing the neuron to go quiet.

2.1 Derivation of the Hindmarsh-Rose Equations

The mathematical foundations of the electrophysics of a single neuron relies on the Hodgkin-Huxley model [1], which can be described by the following differential ordinary equations:

$$\begin{aligned} C \frac{dv}{dt} &= I_{Na} + I_K + I_L + I \\ \frac{dx}{dt} &= \frac{x_\infty(v) - x}{\tau_x(v)}, \quad x = m, h, n \end{aligned} \quad (2.2)$$

where v is the membrane potential, C is the membrane capacity, I is the applied current, and m, h, n are gating variables representing sodium activation, sodium inactivation, and potassium activation respectively.

Because the sodium activation m is significantly faster than the other variables, it is assumed to reach its steady state $m_\infty(v)$ instantaneously. Furthermore, since h and n evolve on similar slow timescales and exhibit a strong linear correlation, they can be collapsed into a single recovery variable. This reduction leads to a two-variable generalization of the Fitzhugh model, characterized by the following differential equations for the action potential:

$$\dot{v} = \frac{1}{C} (r - f(v) + I) \quad (2.3)$$

$$\dot{r} = \gamma (g(v) - \delta r) \quad (2.4)$$

in which v represents the membrane potential, r is the recovery variable (tracking both gating variables), I is the applied current, while C is the membrane capacity, γ and δ are constants.

The form of the functions $f(v)$ and $g(v)$ is determined by voltage clamp experiments and by the requirements that f must be a cubic.

The addition of a slow inward current can be described by the equation

$$\dot{z} = \gamma (h(v) - \delta z) \quad (2.5)$$

then, defining a new variable $w = r + z$, we obtain

$$\dot{w} = \gamma (g(v) + h(v) - \delta w) \quad (2.6)$$

A simple form of the equations (2.3) and (2.6) can be obtained as follows. First transform (2.3) and (2.6) by defining a new time variable $T = \gamma \delta t$ and introducing new variables x and y defined by:

$$\begin{aligned} x(T) &= v(T) \\ y(T) &= \frac{1}{C\gamma\delta} w(t) \end{aligned}$$

which satisfy the differential equations

$$\dot{x} = y - F(x) + \frac{1}{C\gamma\delta} I \quad (2.7)$$

$$\dot{y} = G(x) - y \quad (2.8)$$

Now impose that $F(x)$ is a cubic (and $F \rightarrow \infty$ as $x \rightarrow \infty$), $G(x)$ is a quadratic and both $F(x), G(x)$ have a local maximum for $x = 0$. It yields

$$\dot{x} = y - ax^3 - bx^2 + I \quad (2.9)$$

$$\dot{y} = c - dx^2 - y \quad (2.10)$$

where $a, b, c, d > 0$ are dimensionless parameters. The Slow inward current has the following expression:

$$\dot{z} = r(s(x - x_r) - z) \quad (2.11)$$

The variables are then relabeled as $x_1 = x, x_2 = y, x_3 = z$. The change of variables introduced above defines a natural set of rescaled units for the model. The motivation for this rescaling is to absorb the physical constants C, γ , and δ into the variables, yielding a dimensionless system that is easier to analyze and simulate. The rescaled time $T = \gamma\delta t$ measures time in units of $(\gamma\delta)^{-1}$, the characteristic timescale of the recovery dynamics. The variable x retains the same units as the membrane potential v (millivolts), while the rescaled recovery variable $y = \frac{1}{C\gamma\delta} w$ is dimensionless, having absorbed the membrane capacitance and rate constants. Accordingly, the applied current appears in the rescaled form $\tilde{I} = \frac{1}{C\gamma\delta} I$, and the parameters a, b, c, d that arise from imposing the cubic and quadratic forms are pure dimensionless quantities. All numerical simulations presented in this work are performed in the rescaled system, and therefore all reported timescales are expressed in dimensionless time units T .

2.2 Slow-Fast dynamics

In the limit where $r \rightarrow 0$, the adaptation variable x_3 evolves so slowly that it can be treated as a constant parameter, $x_{3_{const}}$, within the fast dynamics. This allows decoupling the model in a 2D fast subsystem:

$$\begin{cases} \dot{x}_1 &= x_2 - ax_1^3 + bx_1^2 - x_{3_{const}} + I \\ \dot{x}_2 &= c - dx_1^2 - x_2 \end{cases} \quad (2.12)$$

The behavior of the 2D subsystem is fundamental to the generation of the bursting patterns seen in the full 3D model: in this subsystem, x_3 acts as a shift factor for the cubic nullcline. As the slow variable x_3 increases, it effectively moves the cubic nullcline upward, which can cause the stable equilibrium point to vanish or lose stability. The fast subsystem is responsible for the repetitive firing of action potentials. When the fixed point of the 2D system becomes unstable (e.g., via a Hopf bifurcation), the system enters a stable limit cycle, which represents the fast "spiking" phase of a burst. As x_3 continues to grow slowly during the spiking phase, it eventually pushes the system toward a bifurcation point where the limit cycle disappears (often a homoclinic or saddle-node on invariant circle bifurcation), causing the neuron to return to a quiescent state. This behavior is relevant to the bifurcation analysis of the full 3D system presented in the next sections.

Chapter 3

Open loop Analysis

3.1 Equilibrium Points

In order to find the equilibria of the system, one must set the time derivatives equal to zero:

$$\begin{cases} x_2 - ax_1^3 + bx_1^2 - x_3 + I = 0 \\ c - dx_1^2 - x_2 = 0 \\ r(s(x_1 - x_r) - x_3) = 0 \end{cases} \quad (3.1)$$

From the second and third equations of (3.1), we can express x_2 and x_3 as functions of x_1 :

$$x_2 = c - dx_1^2, \quad x_3 = s(x_1 - x_r) \quad (3.2)$$

Substituting these expressions into the first equation yields a single transcendental equation in x_1 :

$$(c - dx_1^2) - ax_1^3 + bx_1^2 - s(x_1 - x_r) + I = 0 \quad (3.3)$$

Rearranging terms and normalizing by the lead coefficient a , we obtain the following cubic equation:

$$x_1^3 + \frac{d-b}{a}x_1^2 + \frac{s}{a}x_1 - \frac{I + sx_r + c}{a} = 0 \quad (3.4)$$

For convenience, this can be rewritten in the standard cubic form:

$$x_1^3 + px_1^2 + qx_1 + k = 0 \quad (3.5)$$

where the coefficients p, q and k are defined by the physical parameters of the system. The nature of these roots is dictated by the cubic discriminant Δ_{eq} , computed by using the cubic discriminant formula (A.8), which is computed as:

$$\Delta_{eq} = 18pqr - 4p^3k + p^2q^2 - 4q^3 - 27k^2 \quad (3.6)$$

Consistent with the properties of cubic polynomials, $\Delta_{eq} > 0$ indicates three distinct real roots, while $\Delta_{eq} < 0$ indicates a single real root. For any given set of parameters, the number of equilibrium points is determined by the roots of (3.5):

- **One real root** and two complex conjugate roots: The system possesses **one equilibrium point**.
- **Three real roots**: The system possesses **three equilibrium points**.

3.2 Stability analysis of equilibria

The stability of the equilibria is investigated using linear approximation. The analytical expression for the Jacobian matrix $\mathbf{J}$ at a fixed point (x_1^*, x_2^*, x_3^*) is defined as:

$$\mathbf{J}(x_1^*) = \begin{bmatrix} -3ax_1^{*2} + 2bx_1^* & 1 & -1 \\ -2dx_1^* & -1 & 0 \\ rs & 0 & -r \end{bmatrix} \quad (3.7)$$

At an equilibrium point, the stability is governed by eigenvalues λ , which are the roots of the characteristic polynomial:

$$P(\lambda) = \det(\mathbf{J} - \lambda\mathbf{I}) = 0 \quad (3.8)$$

Expanding the determinant, it yields:

$$P(\lambda) = \lambda^3 + \alpha_2\lambda^2 + \alpha_1\lambda + \alpha_0 = 0 \quad (3.9)$$

where the coefficients are defined as:

$$\begin{aligned} \alpha_2 &= -3ax_1^{*2} + 2bx_1^* + 1 + r \\ \alpha_1 &= (-3ax_1^{*2} + 2bx_1^*)(1 + r) + r + 2dx^* + rs \\ \alpha_0 &= r(-3ax_1^{*2} + 2bx_1^* + 2dx^* + rs) \end{aligned} \quad (3.10)$$

The discriminant of the characteristic polynomial is

$$\Delta = 18\alpha_2\alpha_1\alpha_0 - 4\alpha_2^3\alpha_0 + \alpha_2^2\alpha_1^2 - 4\alpha_1^3 - 27\alpha_0^2 \quad (3.11)$$

The discriminant tells us about the nature of the roots, so the nature of the eigenvalues:

- $\Delta > 0$ Three distinct real eigenvalues.
- $\Delta = 0$ Eigenvalues with multiplicity greater than one.
- $\Delta < 0$ One real eigenvalue and a complex conjugate pair.

Then to establish the stability associated with those eigenvalues, the Routh-Hurwitz condition must be applied:

$$\begin{aligned} \alpha_2 &> 0 \\ \alpha_1 &> 0 \\ \alpha_0 &> 0 \\ \alpha_2\alpha_1 &> \alpha_0 \end{aligned} \quad (3.12)$$

When these conditions hold, all eigenvalues have negative real parts. The cases are analysed separately as follows:

- **Case I:** $\alpha_2 > 0, \alpha_1 > 0, \alpha_0 > 0, \alpha_2\alpha_1 > \alpha_0$. All eigenvalues have negative real parts.
 - $\Delta > 0$ Three distinct real eigenvalues, so a Stable Node.
 - $\Delta < 0$ One real eigenvalue and a pair of complex conjugate, so a Stable focus.
- **Case II:** At least one condition fails.
 - $\Delta > 0$ Three distinct real eigenvalues, so an unstable Node or a saddle.
 - $\Delta < 0$ One real eigenvalue and a pair of complex conjugate, so an unstable focus or a saddle.

3.3 Bifurcation analysis

The system is analyzed using the parameters listed in Table 3.1. This study reports the results of a systematic investigation into the dynamical behaviors induced by the modification of the external current I , varied in the range $[0.0, 3.5]$. While I serves as the primary bifurcation parameter, further analysis could be extended to parameters such as r or b .

Parameter	Value
a	1
b	3
c	1
d	5
r	0.003
s	4
x_r	-1.6
I	$[0.0, 3.5]$

Table 3.1: Parameter table: Bifurcation analysis

In this context, considering I as the independent variable, the equilibrium equation (3.4) is expressed as:

$$p(x) = x_1^3 + 2x_1^2 + 4x_1 + \left(\frac{27}{5} - I\right) = 0 \quad (3.13)$$

The real root, $x_1^*(I)$, determines the remaining coordinates of the fixed point according to:

$$x_2^*(I) = 1 - 5x_1^{*2}(I), \quad x_3^*(I) = 4x_1^*(I) + \frac{32}{5}$$

The local stability of this equilibrium is dictated by the eigenvalues of the Jacobian matrix (3.7):

$$\mathbf{J}(x_1^*) = \begin{bmatrix} -3x_1^{*2} + 6x_1^* & 1 & -1 \\ -10x_1^* & -1 & 0 \\ 0.012 & 0 & -0.003 \end{bmatrix} \quad (3.14)$$

where the numerical values for the third row are derived from the nominal parameters $r = 0.003$ and $s = 4$. The eigenvalues are the roots of the characteristic polynomial $P(\lambda) = \lambda^3 + \alpha_2(I)\lambda^2 + \alpha_1(I)\lambda + \alpha_0(I)$, with coefficients defined as:

$$\begin{aligned} \alpha_2(I) &= 1.003 - 6x_1^*(I) + 3x_1^{*2}(I) \\ \alpha_1(I) &= 0.003(5 - 6x_1^*(I) + 3x_1^{*2}(I)) + 4x_1^*(I) + 3x_1^{*2}(I) \\ \alpha_0(I) &= 0.003(4 + 4x_1^*(I) + 3x_1^{*2}(I)) \end{aligned} \quad (3.15)$$

As the input current I varies, the equilibrium coordinates shift and the eigenvalues traverse the complex plane. Characterizing the system's transition between regimes requires an investigation into the conditions for a Hopf bifurcation.

Numerical verification through *matcont* identifies a **subcritical Hopf bifurcation** at $I \approx 1.305635$ and a **Limit Point of Cycles (LPC)** bifurcation at $I \approx 1.2762302$.

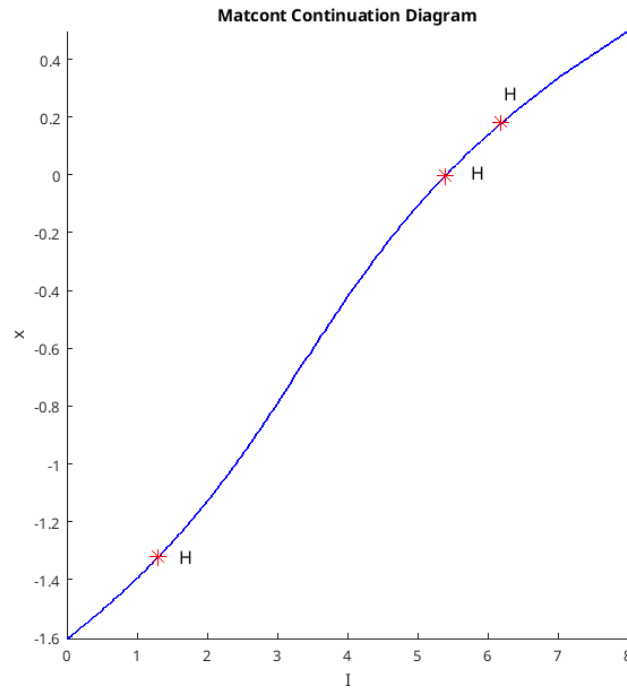


Figure 3.1: Matcont continuation diagram showing the evolution of the equilibrium branch and bifurcation points.

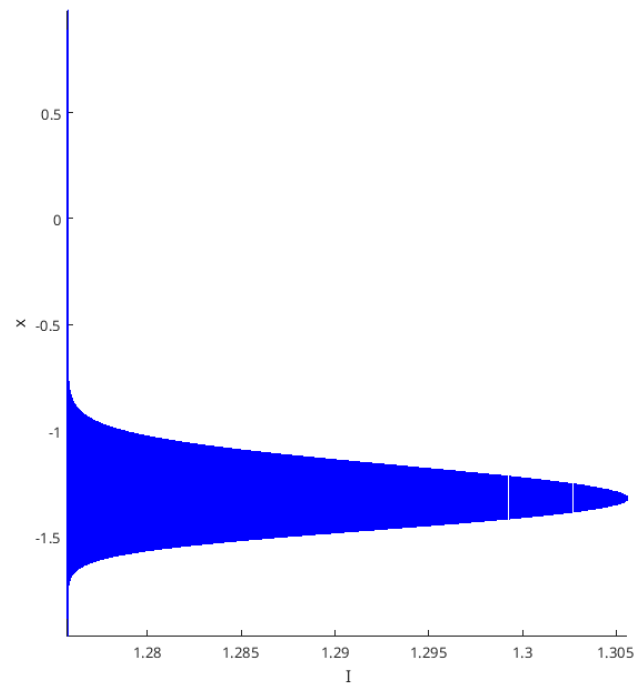


Figure 3.2: Continuation of the limit cycle branch identifying the LPC bifurcation.

For low values of input current, particularly before $I \approx 1.276$, the system exhibits a single global attractor represented by a **stable focus**. In this regime, all trajectories,

regardless of initial conditions, converge to the equilibrium point, corresponding to the biological "resting state" or quiescent state of the neuron, as shown in Figure (3.4).

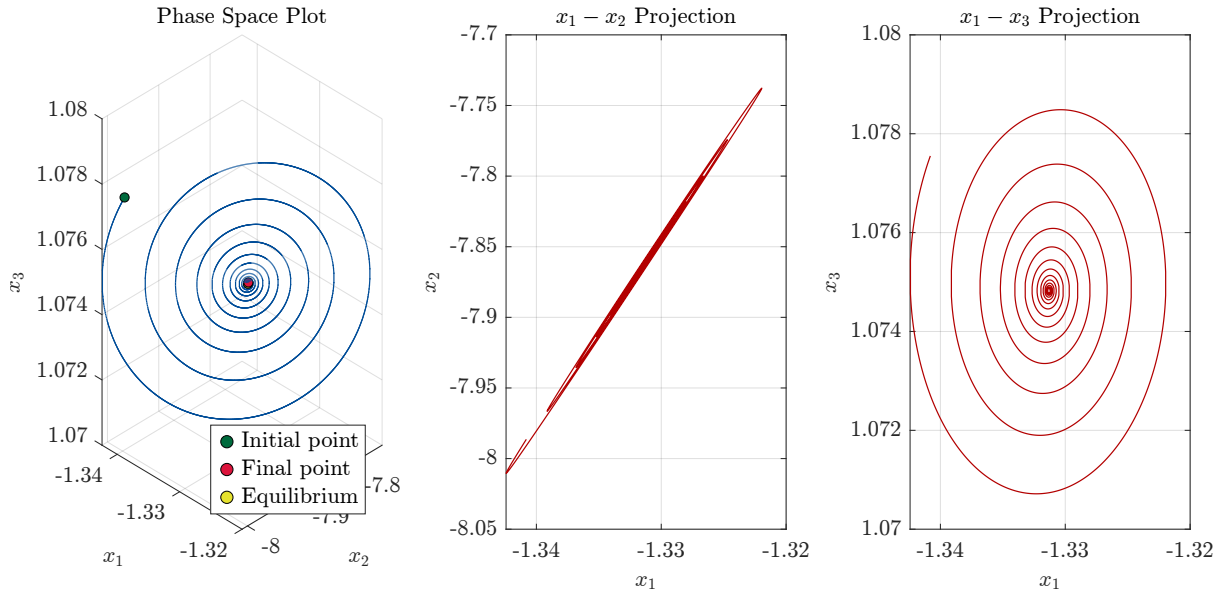


Figure 3.3: Phase space analysis: stable focus

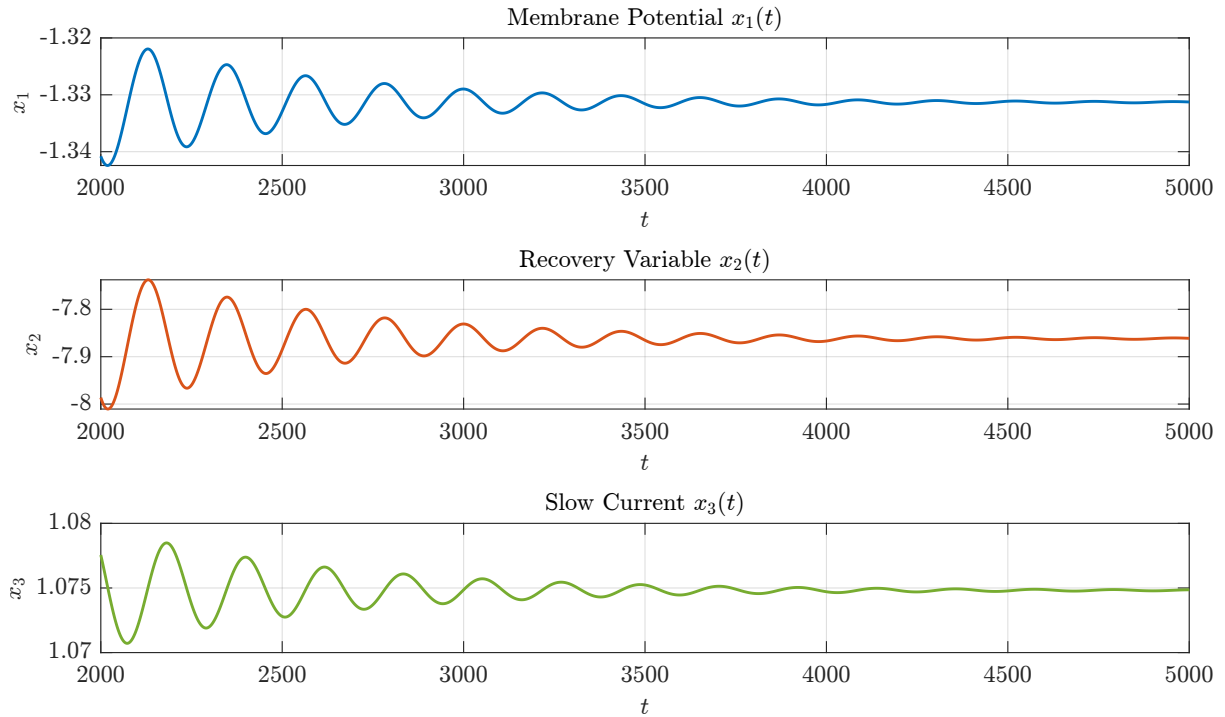


Figure 3.4: Time Series analysis: resting state

At $I \approx 1.276$, a Limit Point of Cycles (LPC) bifurcation occurs. This bifurcation marks the appearance of two limit cycles with a non-zero amplitude:

- **Stable Limit Cycle:** A large orbit representing the spiking phenomenon.
- **Unstable Limit Cycle:** A "protection barrier" that separates the basins of attraction between the stable focus and the stable limit cycle.

This creates a region of **bistability**. As shown in the simulations, the system can either jump to the periodic orbit as in Figure 3.5 (if the "threshold" of the unstable cycle is crossed) or remain at the stable focus as in Figure 3.7 (if perturbations are small).

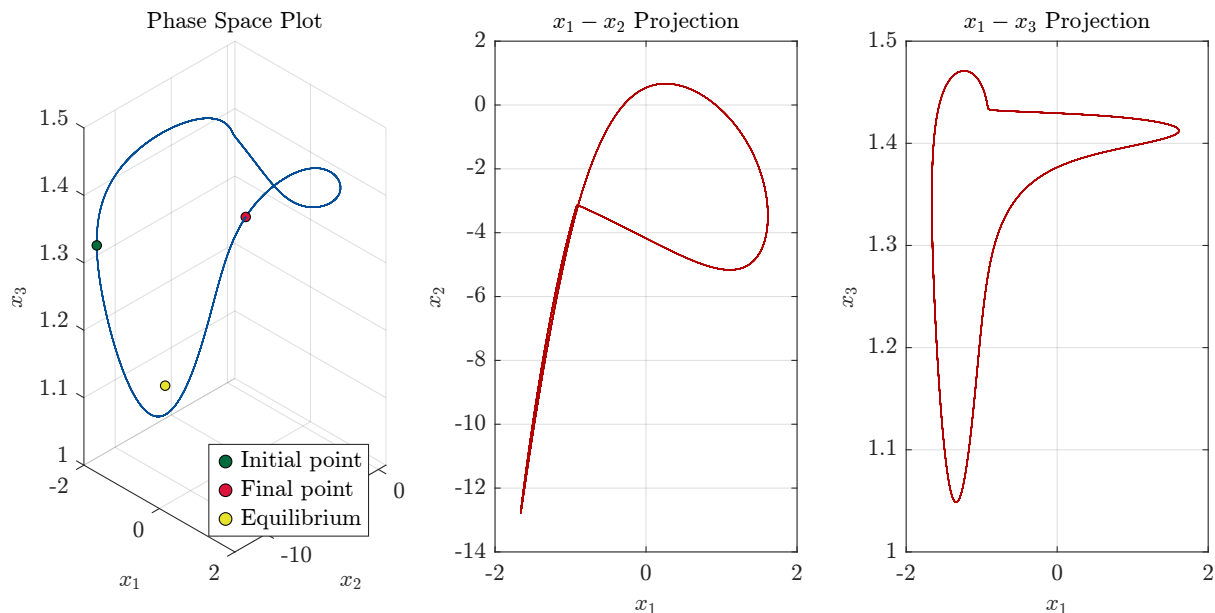


Figure 3.5: Phase space analysis: stable orbit

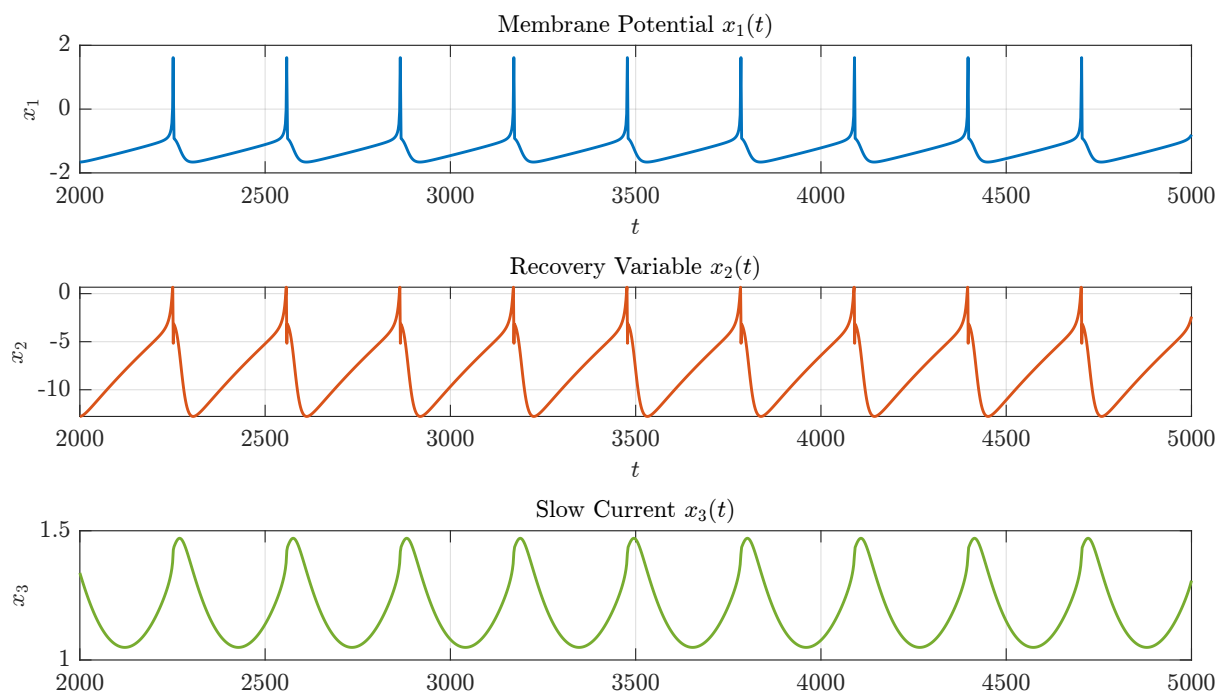


Figure 3.6: Time Series analysis: spiking

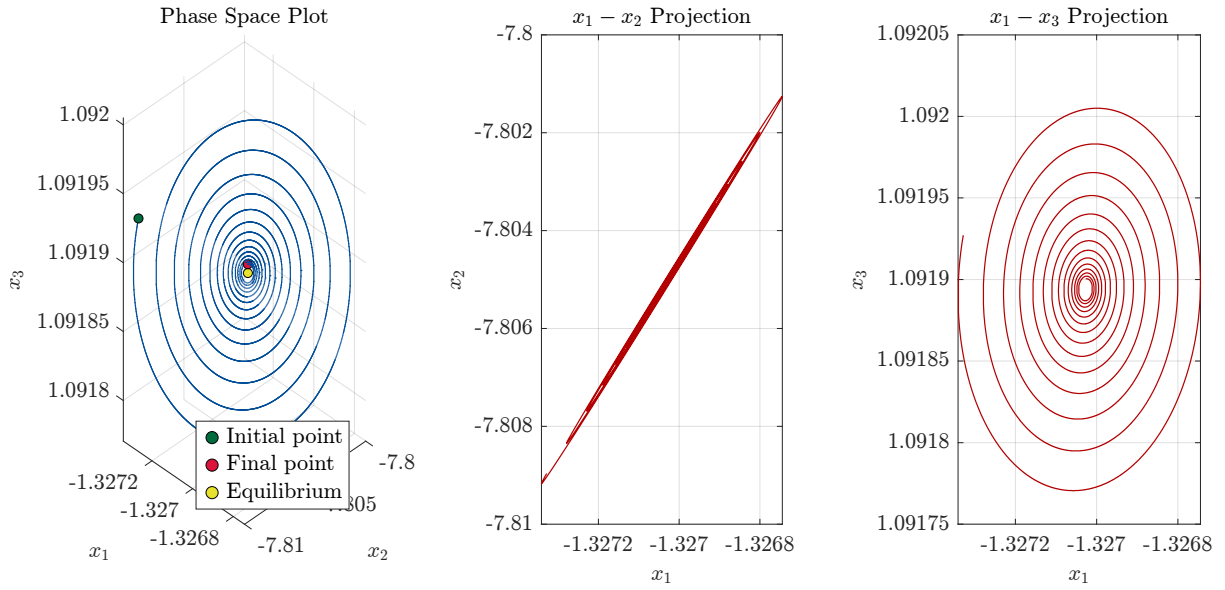


Figure 3.7: Phase space analysis: stable focus

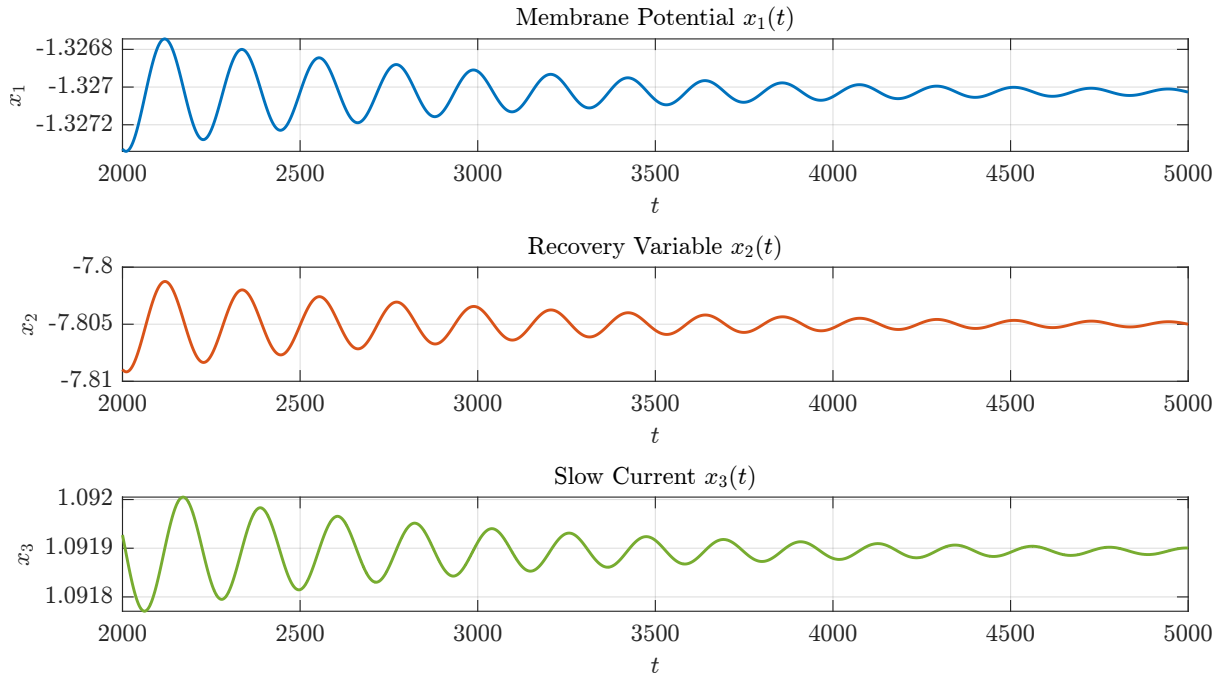


Figure 3.8: Time Series analysis: stable focus

As the input current I is increased further, the periodic attractor undergoes a series of secondary bifurcations. Specifically, at $I \approx 1.302$, the system exhibits a folding of the phase space trajectory (Figure 3.9). This phenomenon characterizes a **period-adding bifurcation**, where the dynamics transition from simple period-1 spiking into a period-2 bursting pattern (Figure 3.10). The phase space plots illustrate the trajectory "folding" over itself, which serves as a critical precursor to the emergence of chaotic dynamics and block structured patterns observed in the Hindmarsh-Rose model.

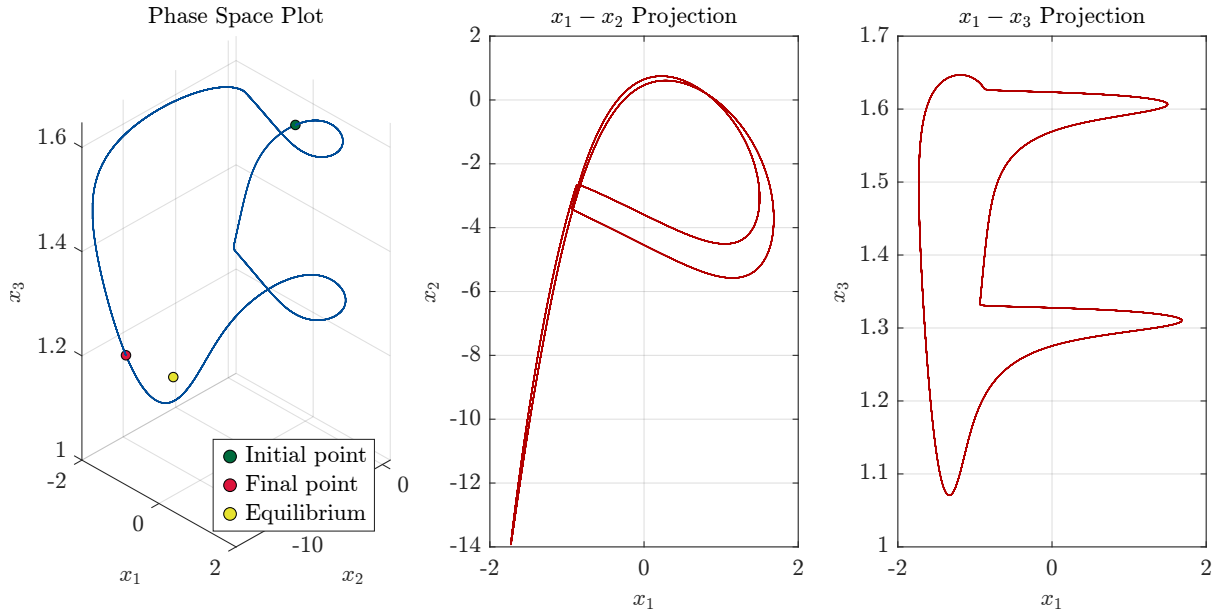


Figure 3.9: Phase space analysis: Period-Adding bifurcation

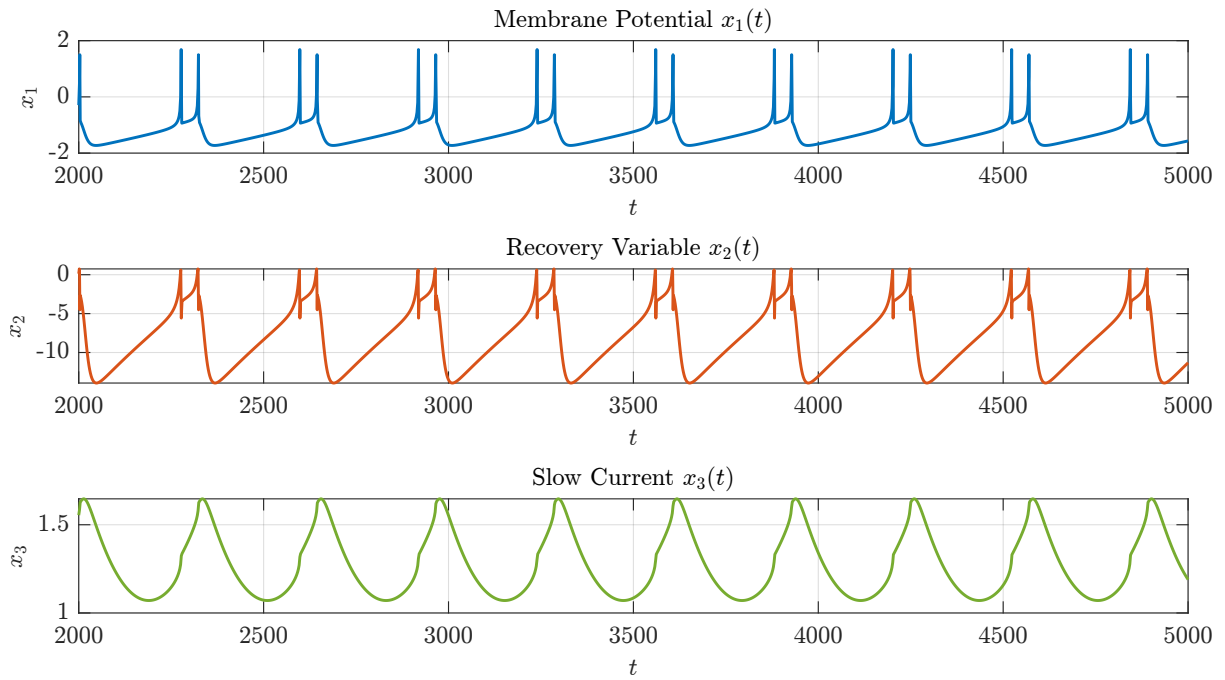


Figure 3.10: Time Series analysis: Bursting phenomenon

The resting state eventually loses stability via the **subcritical Hopf bifurcation** at $I \approx 1.3056$. At this point:

- The First Lyapunov Coefficient (l_1) is positive, confirming the subcritical nature of the bifurcation.
- The unstable limit cycle collides with the stable focus, leaving behind an unstable saddle-focus, as shown in figure (3.11).
- After this point, the equilibrium is no longer an attractor: all trajectories are forced toward the global spiking/bursting attractor (figure 3.13).

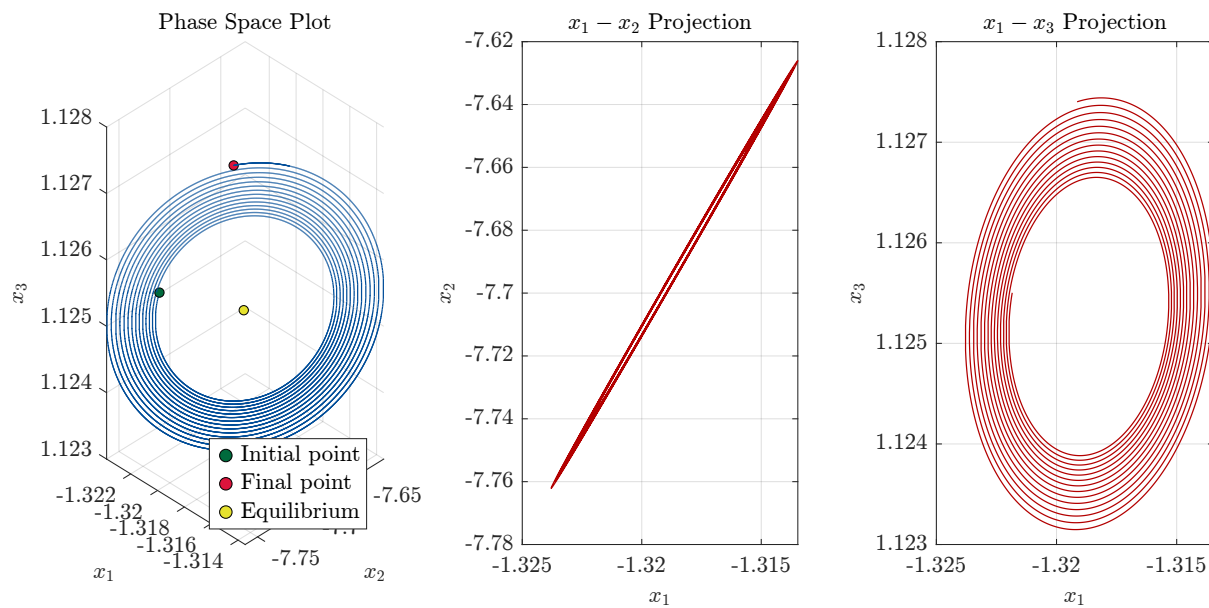


Figure 3.11: Phase space analysis: Unstable Saddle-Focus after the Subcritical Hopf bifurcation

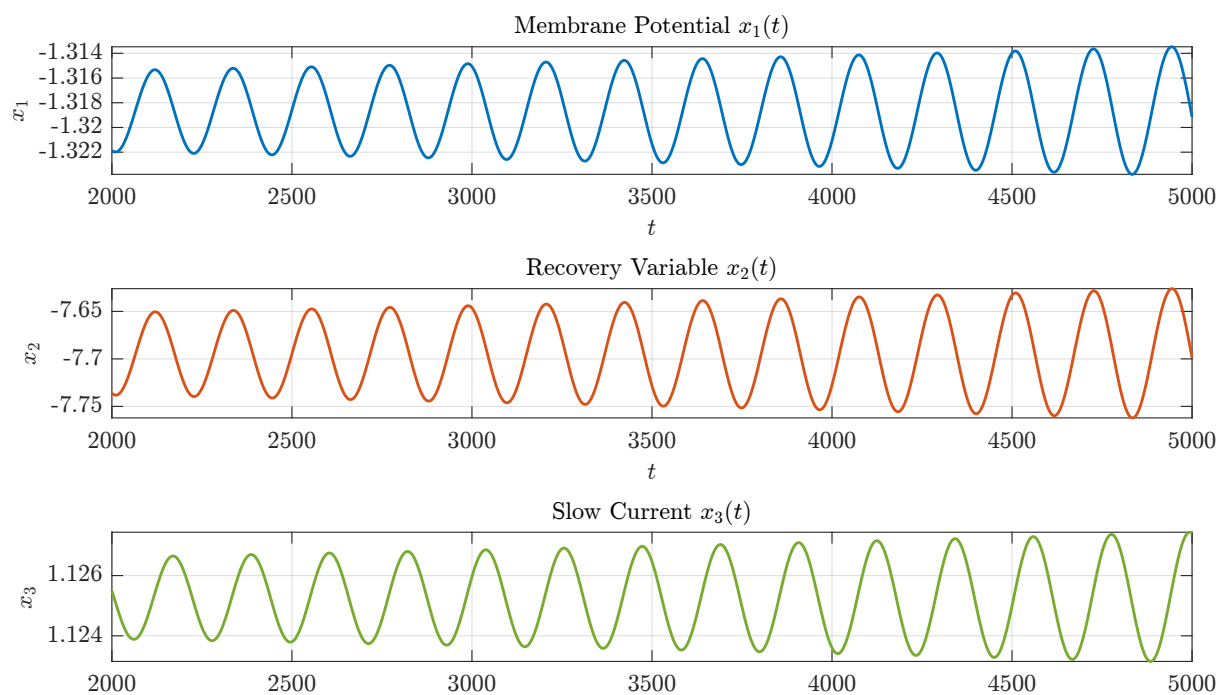


Figure 3.12: Time Series analysis: Unstable behavior after the Subcritical Hopf bifurcation

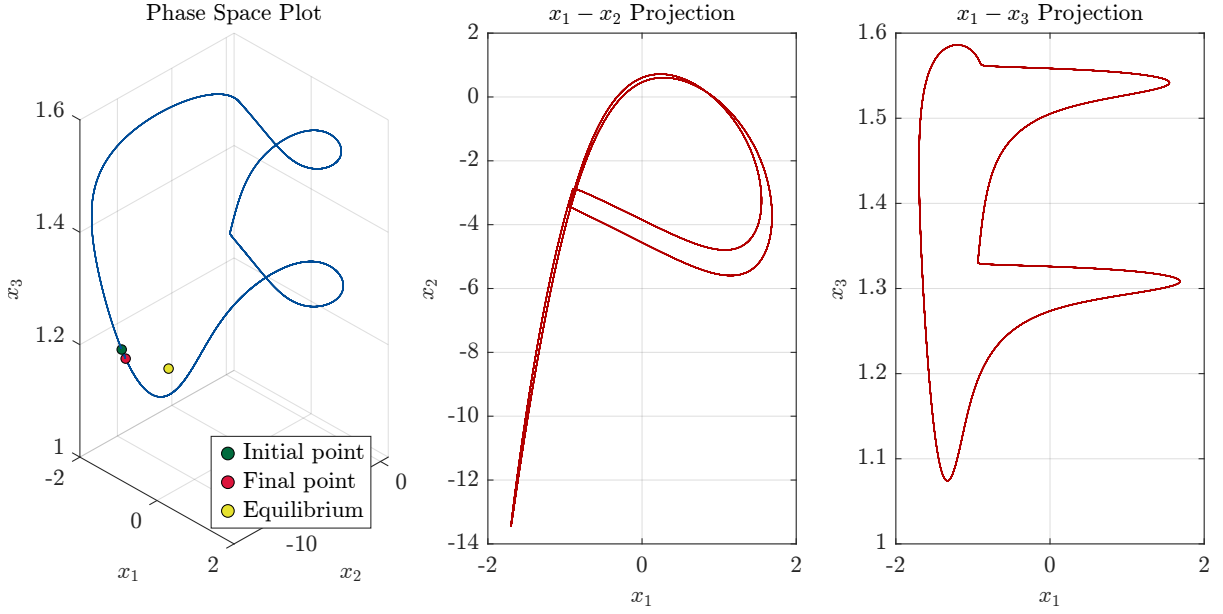


Figure 3.13: Phase space analysis: Bursting

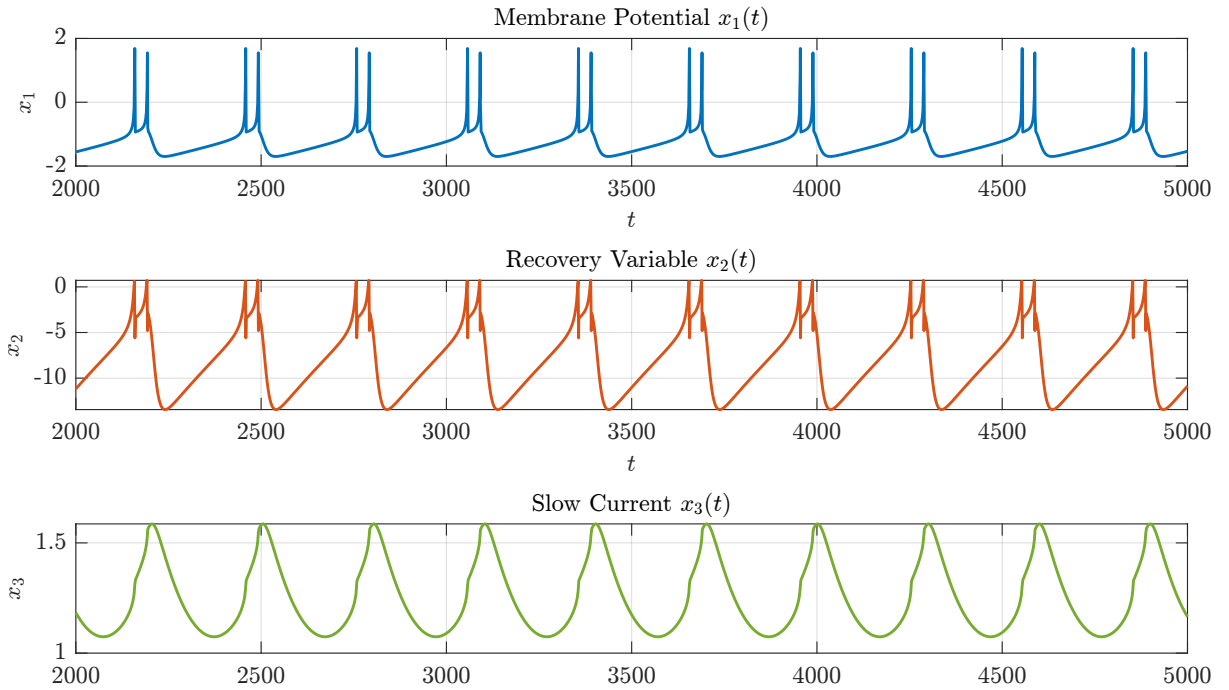


Figure 3.14: Time series analysis: Bursting

3.4 Complex Bifurcation Structures and Chaos

In this section, the global transition from periodic spiking to chaotic bursting is quantitatively analyzed through the Inter-Spike Interval (ISI) diagram (figure 3.15) and a Poincaré mapping calculated at the section $x_1 = 0$ (figure 3.16). Following the work of González-Miranda [2], these diagrams characterize the system's sensitivity to the bifurcation parameter I through the following behaviors:

- Period-Adding Cascade** ($1.28 \leq I \leq 3.10$): Shortly after the LPC bifurcation at $I \approx 1.276$, the period-adding cascade becomes clearly observable. In this range, the ISI evolves in a staircase-like fashion, where the dynamical behavior is partitioned into discrete blocks. The periodicity p (the number of spikes per burst) increases incrementally with I . Notably, the window for simple periodic spiking ($p = 1$) is confined to a narrow interval $I \in [1.28, 1.30]$, providing numerical evidence of the proximity between the LPC and subcritical Hopf bifurcations.
- Crisis and Chaos** ($3.10 < I < 3.35$): A significant topological shift occurs as the parameter approaches the critical value $I_C \approx 3.31$. The dense clouds of points observed in both Figure 3.15 and Figure 3.16 indicate a transition to chaotic bursting via a sequence of period-doubling events. At I_C , a sharp change in the width of the bifurcation diagram reveals a **Boundary Crisis**, where the chaotic attractor collides with an unstable periodic orbit, leading to a sudden expansion of the chaotic set (Figure 3.17).
- Reverse Period-Doubling and Tonic Spiking** ($I > 3.35$): To the right of the chaotic window, the system undergoes an **Inverted Period-Doubling Cascade**. On the side where the diagram narrows, the multiple branches of the Poincaré map converge into a single stable branch. This marks the recovery of regular **tonic spiking** dynamics, where the slow adaptation variable x_3 can no longer suppress the membrane potential x_1 sufficiently to create silent phases, resulting in a continuous firing pattern.

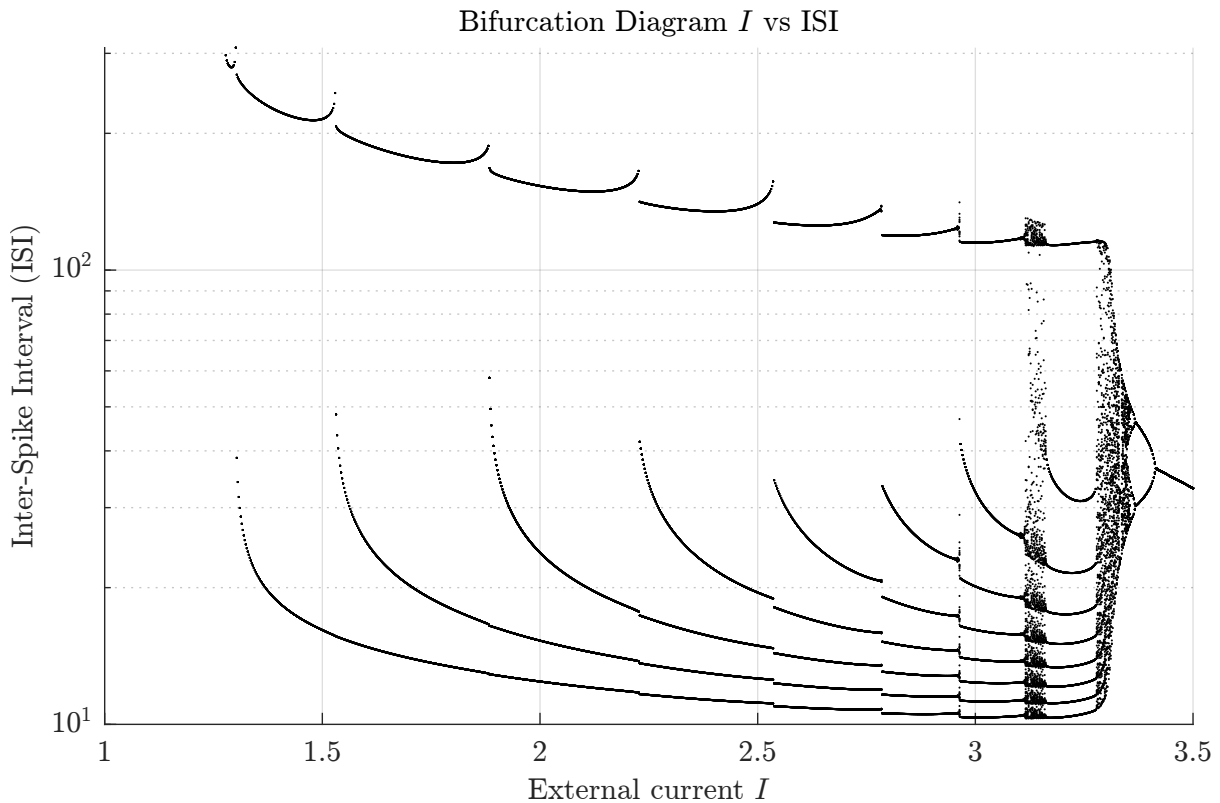


Figure 3.15: Bifurcation Diagram: Inter-Spike Interval vs External Current I

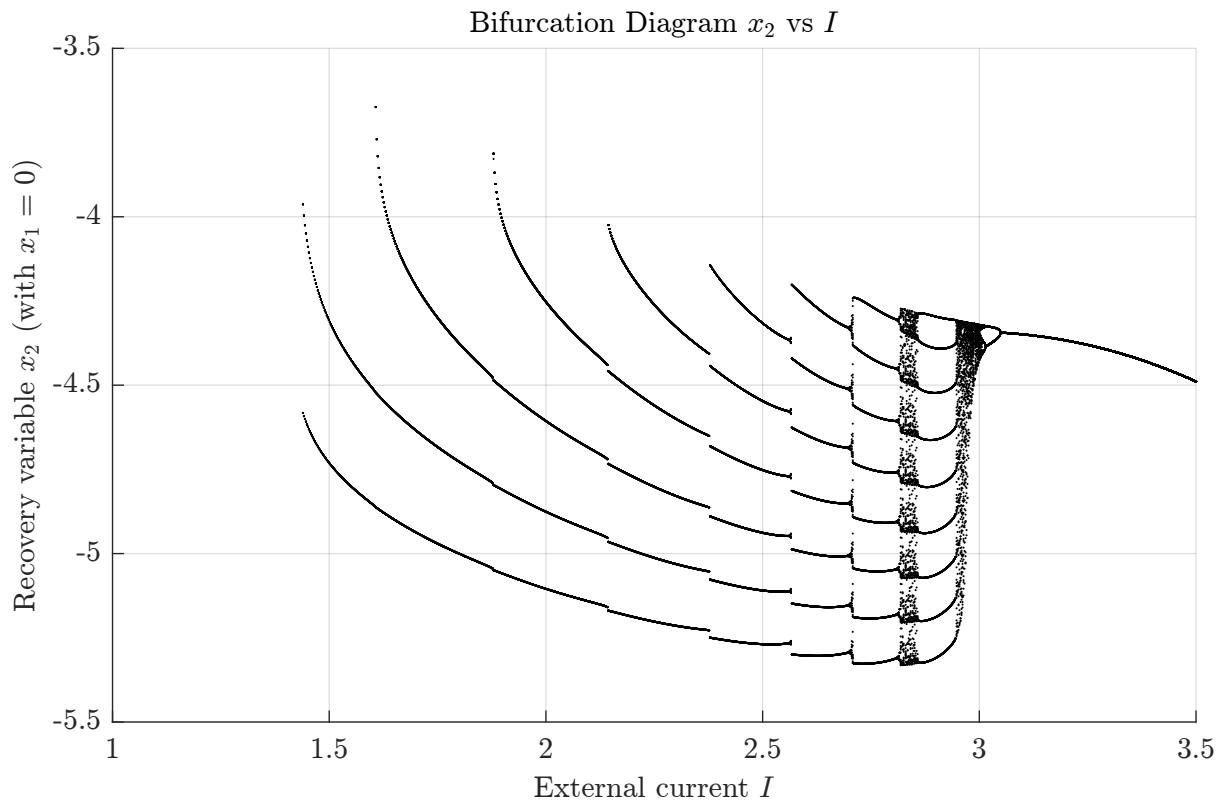
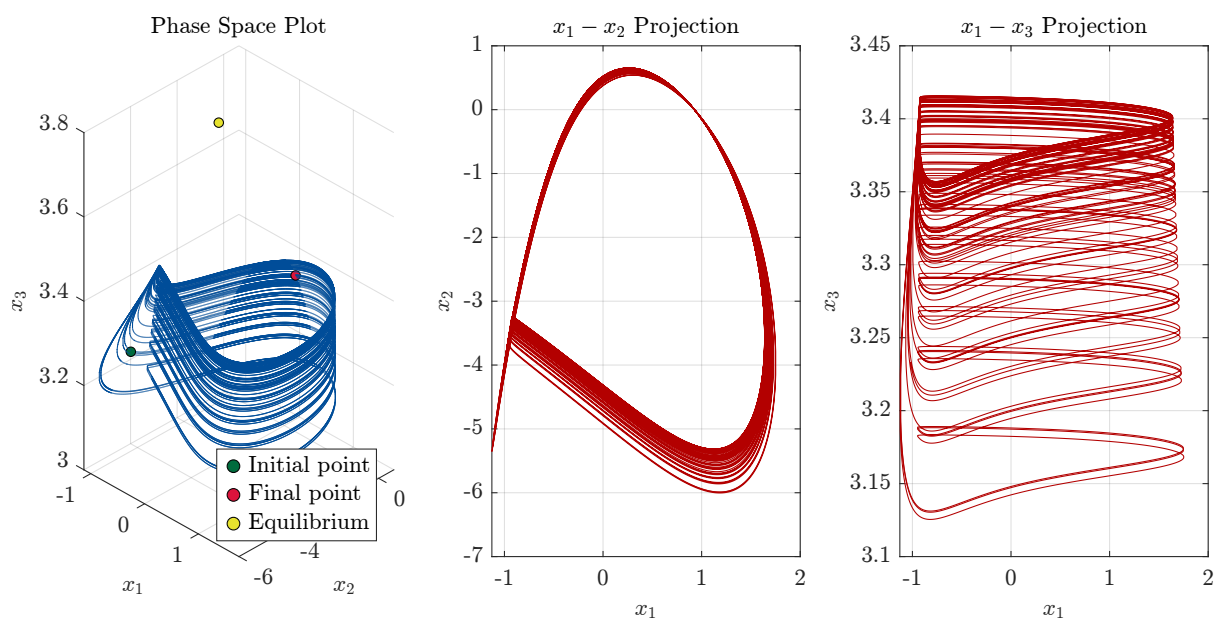
Figure 3.16: Bifurcation Diagram: Poincaré section $x = 0$ vs External Current I 

Figure 3.17: Chaotic attractor: phase space representation

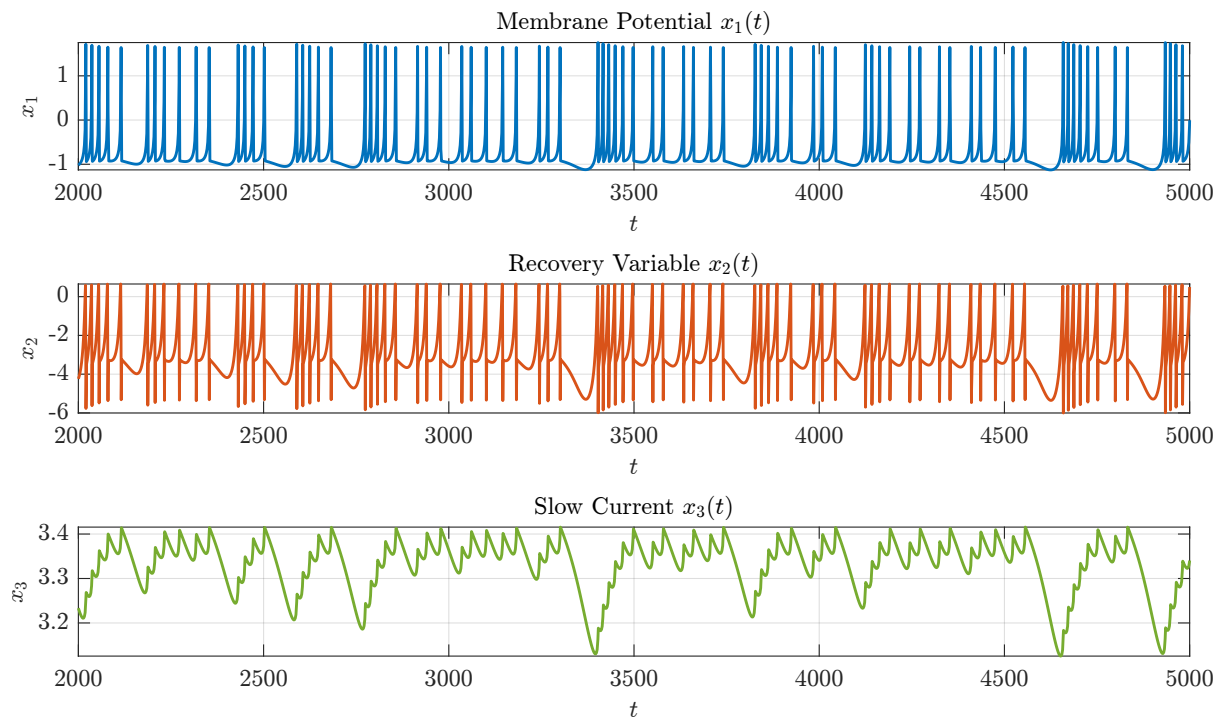


Figure 3.18: Chaotic attractor: time series representation

All plots were produced using the *MATLAB* scripts `flow.m` and `bifurcation_diagram.m`.

Chapter 4

Synthesis

From this point forward, the Hindmarsh-Rose (HR) system is analyzed using the parameters defined in Table 4.1 and Table 4.2. These values establish the nominal configuration for the design and performance evaluation of the proposed control strategies. The dynamical model is defined as:

$$\begin{cases} \dot{x}_1 &= x_2 - ax_1^3 + bx_1^2 - x_3 + I + u \\ \dot{x}_2 &= c - dx_1^2 - x_2 \\ \dot{x}_3 &= r[s(x_1 - x_r) - x_3] \end{cases} \quad (4.1)$$

where $\mathbf{x} = [x_1, x_2, x_3]^\top \in \mathbb{R}^3$ represents the state vector. Here, x_1 denotes the membrane potential, x_2 is the recovery variable (associated with fast ion currents), and x_3 represents the slow adaptation current. The **control input** u represents an independent external current injected into the neuron. Unlike the constant bias current I , which dictates the neuron's intrinsic firing regime (e.g., quiescent, spiking, or bursting), u simulates the intervention of a neurostimulation device, such as a pacemaker or a Deep Brain Stimulation (DBS) electrode, rather than an endogenous synaptic signal.

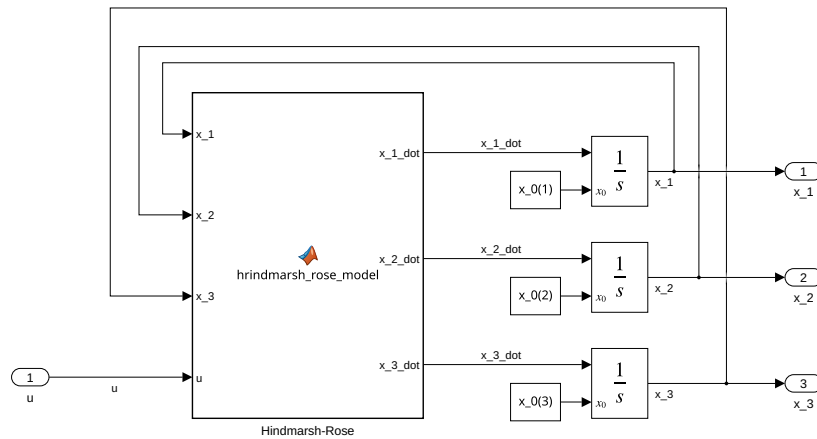


Figure 4.1: Simulink implementation (inside the mask) of the Hindmarsh-Rose Dynamics

The proposed control strategies are designed to satisfy two primary technical objectives:

Parameter	Value
a	1
b	3
c	1
d	5
r	0.003
s	4
x_r	-1.6
I	1.28

Table 4.1: Nominal parameters for regulation tasks (Spiking regime)

- **Setpoint Regulation:** Stabilizing the system at a desired equilibrium $\mathbf{x}^* = [-1.60, -11.87, -0.01]^\top$, effectively transitioning the neuron from a spiking regime to a quiescent state, assuming the open-loop neuron is initially in a spiking regime.
- **Trajectory Tracking:** Forcing the system state $\mathbf{x} = [x_1, x_2, x_3]^\top$ to follow a time-varying spiking reference signal $x_{ref}(t)$ stimulated by a leader neuron in a spiking regime, assuming the open-loop neuron is initially in a quiescent state. Note that the tracking problem is similar to a **pinning control** technique, where a 'leader' neuron dictates the trajectory and the 'follower' neuron utilizes the control action to achieve synchronization.

The initial conditions for the regulation task are set to $\mathbf{x}_0 = [0.1, 0.1, 0.1]^\top$, which lies outside the basin of attraction of the stable focus, ensuring the controller must actively drive the trajectory toward the desired equilibrium. For the tracking task, the initial conditions are set to $\mathbf{x}_0 = [-1.6, -11.87, 0]^\top$, corresponding to the quiescent equilibrium of the neuron with $I = 0.0$.

Parameter	Value
a	1
b	3
c	1
d	5
r	0.003
s	4
x_r	-1.6
I	0.0

Table 4.2: Nominal parameters for tracking tasks (Quiescent regime)

In the following, multiple control strategies are investigated:

- **Classical PID Control:** A linear feedback mechanism utilized to provide a performance baseline.
- **Input Output Feedback Linearization control (FBL):** A nonlinear approach that employs an algebraic transformation to cancel the system's nonlinear dynamics, allowing for the design of a controller in a linear space.

- **Sliding Mode Controller (SMC):** A robust nonlinear technique designed to ensure stability and tracking precision despite potential model mismatches or external perturbations.

4.1 Control specifications

To evaluate and compare the effectiveness of the proposed control strategies, the following performance specifications have been defined for the regulation task, as detailed in Table 4.3

Metric	Definition	Target Value
T_s	Settling time within 2% of x^*	< 5
S_p	Maximum percentage overshoot	$< 5\%$
e_{ss}	Max. steady-state error	< 0.01
u_{max}	Max. admissible control current	± 10

Table 4.3: Performance specifications for the stabilization task.

Since all simulations are performed in the rescaled dimensionless system, all quantities are expressed in dimensionless units. For the trajectory tracking task, the performance is assessed based on the accuracy and precision of the state following the reference signal, as detailed in Table 4.4. To focus on the synchronization capability of the controllers and

Metric	Definition	Target Value
RMSE	Root Mean Square Error	< 0.05
e_{max}	Max. instantaneous tracking error	< 0.20

Table 4.4: Performance specifications for the tracking task.

mitigate the influence of initial condition mismatches, these tracking metrics are evaluated after a transient period of 500 time units. Before using each *SIMULINK* model, please run `params.m`.

4.2 PID Control

To implement the PID control strategy, the nonlinear Hindmarsh-Rose system is first linearized in the neighborhood of the desired equilibrium point. Following linearization, the controller parameters were tuned using a trial-and-error approach, guided by the root locus and frequency response methods, to satisfy the previously defined performance requirements. The Simulink implementation of the control architecture is depicted in Figure 4.2 and implemented in `hr_PID.slx`.

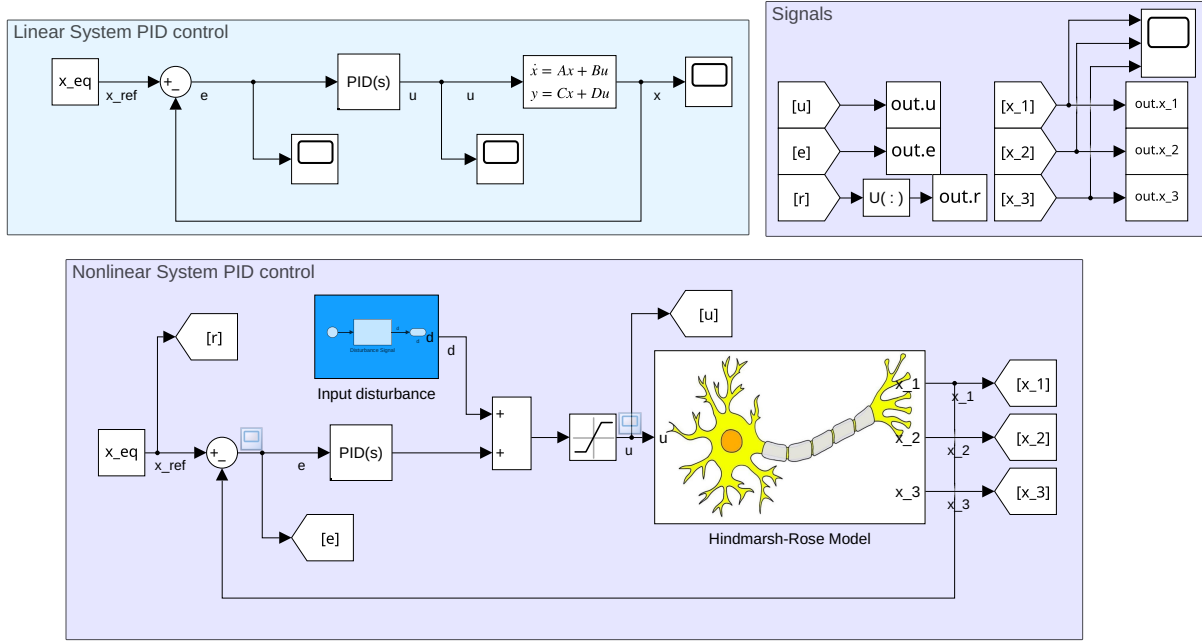


Figure 4.2: Simulink implementation of the PID control for the Hindmarsh-Rose model.

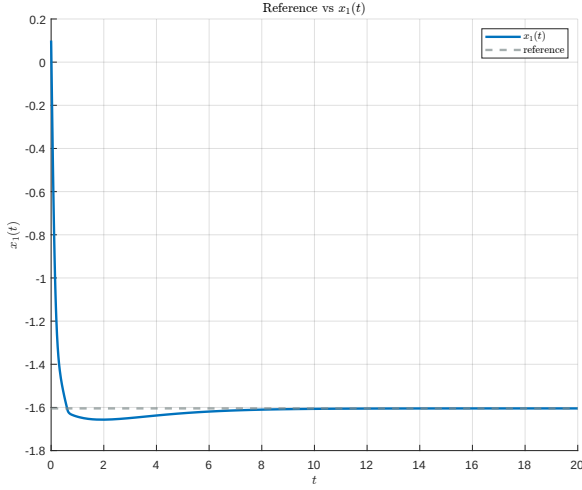
4.2.1 Regulation

The system was linearized around the desired equilibrium point $\mathbf{x}^*$ using a first-order Taylor series expansion. The linearized dynamics are governed by the Jacobian matrix:

$$J = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}^*, u^*} \quad (4.2)$$

By evaluating the partial derivatives at the equilibrium, we obtain the state-space representation $\delta \dot{\mathbf{x}} \approx J \delta \mathbf{x} + B \delta u$, where $\delta \mathbf{x}$ represents small perturbations around the equilibrium. This linear approximation provides the transfer function necessary to evaluate the system's open-loop stability and estimate the initial controller gains.

The PID gains (K_P, K_I, K_D) were subsequently refined to reduce overshoot and settling time while ensuring zero steady-state error. The resulting regulation performance is illustrated in Figure 4.3, where an additional plot demonstrates the stability of the internal state variables over an extended time horizon.



(a) Regulation performance

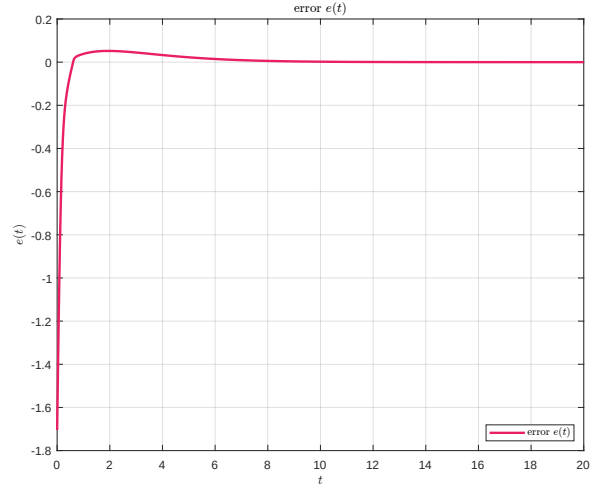
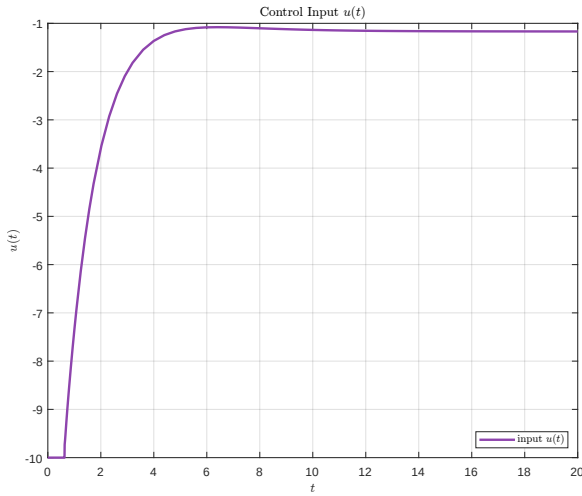
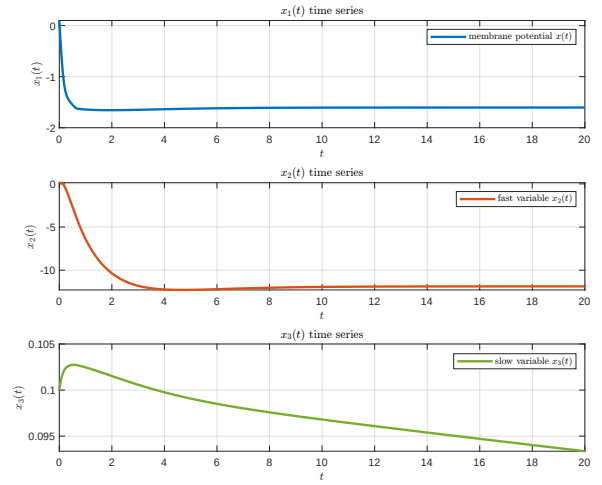
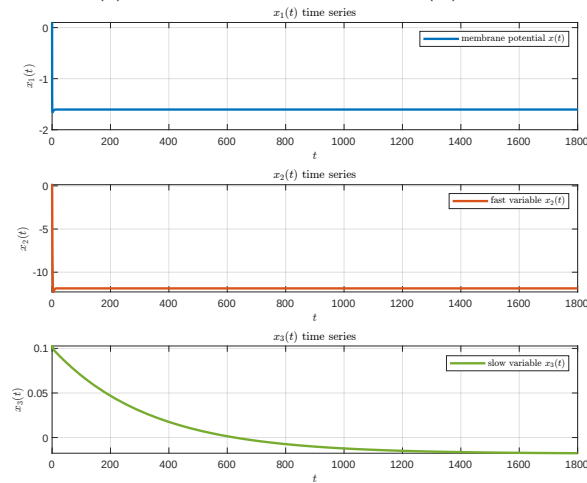
(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) State variables (x_1, x_2, x_3) (e) State variables (extended) (x_1, x_2, x_3)

Figure 4.3: PID regulation results under nominal conditions: (a) membrane potential response, (b) regulation error, (c) control effort, and (d-e) evolution of state variables.

The design specifications are met, using $K_P = 108.97$, $K_I = 49.48$ and $K_D = 2.67$. The system achieves a settling time of $T_s = 3.59$ and an overshoot of $S_p = 3.05\%$.

4.3 Input-Output Feedback Linearization

The first nonlinear control architecture investigated is **Input-Output Feedback Linearization (FBL)**, addressed in the file `hr_io_feedback_linearization.slx`. This method employs a change of variables and a nonlinear control law to transform the nonlinear neuron dynamics into a linear system. The Simulink implementation of the control architecture is depicted in Figure 4.4.

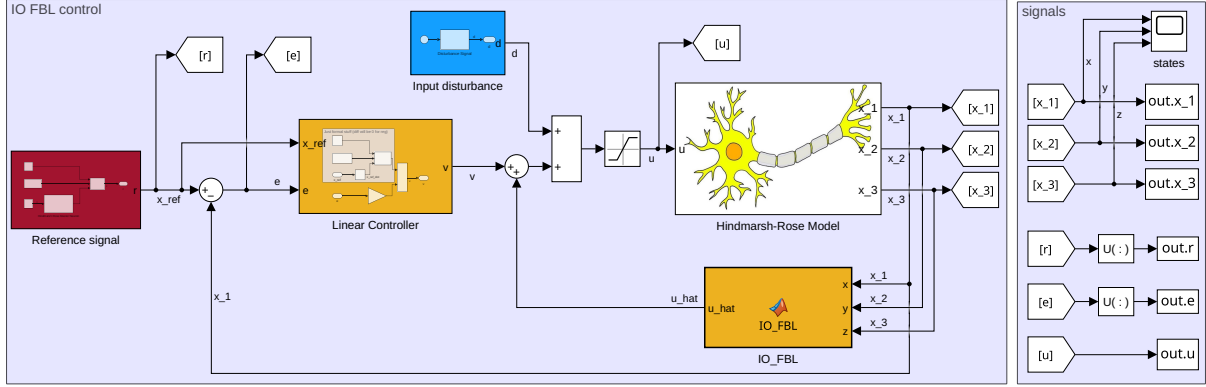


Figure 4.4: Simulink implementation of the Input-Output Feedback Linearization for the Hindmarsh-Rose model.

The design procedure consists of the following steps:

- **Find the Relative Degree:** To determine the relative degree r , we evaluate the Lie derivatives of the output function $h(\mathbf{x}) = x_1$ along the vector fields $f(\mathbf{x})$ and $g(\mathbf{x})$. Considering that the control input u enters the first state equation, one has:

$$\mathcal{L}_g h(\mathbf{x}) = \nabla h \cdot \mathbf{g}(\mathbf{x}) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1 \quad (4.3)$$

Since $\mathcal{L}_g h(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in \mathbb{R}^3$, the system possesses a relative degree $r = 1$. Because $r < n$ (where $n = 3$), there exists an internal dynamic of dimension $n - r = 2$.

- **Feedback linearizing Control law:** We define a nonlinear control law to cancel the system's natural nonlinearities and impose a linear relationship between a new auxiliary input v and the output $h(\mathbf{x})$. The control law is formulated as:

$$u = \frac{1}{\mathcal{L}_g h(\mathbf{x})} [-\mathcal{L}_f h(\mathbf{x}) + v] \quad (4.4)$$

Substituting the Hindmarsh-Rose dynamics, where $\dot{x} = f_1(\mathbf{x}) + u$, the law simplifies to:

$$u = -f_1(\mathbf{x}) + v \quad (4.5)$$

The complete feedback linearizing control law for the Hindmarsh-Rose model is:

$$u = -(x_2 - ax_1^3 + bx_1^2 - x_3 + I) + v \quad (4.6)$$

This results in the decoupled linear dynamics:

$$\dot{h}(\mathbf{x}) = \dot{x}_1 = v \quad (4.7)$$

- **Check stable internal dynamics:** Since $r < n$, we must ensure the stability of the zero dynamics to guarantee that the state trajectories remain bounded. By enforcing the output and its derivatives to be zero ($x_1 = 0, \dot{x}_1 = 0$), the internal states (x_2, x_3) evolve according to:

$$\begin{cases} \dot{x}_2 = c - dx_1^2 - x_2 \\ \dot{x}_3 = r[s(x_1 - x_r) - x_3] \end{cases} \quad (4.8)$$

Evaluating this at the output constraint $x_1 = 0$, the zero dynamics become:

$$\begin{cases} \dot{x}_2 = c - x_2 \\ \dot{x}_3 = r(-sx_r - x_3) \end{cases} \quad (4.9)$$

This subsystem is affine and linearly stable. The Jacobian of the zero dynamics is given by:

$$J_{ZD} = \begin{bmatrix} -1 & 0 \\ 0 & -r \end{bmatrix} \quad (4.10)$$

The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -r$. Since $r > 0$ in the Hindmarsh-Rose parameters, the equilibrium point $(c, -sx_r)$ is a globally asymptotically stable node, confirming that the input-output linearization is internally stable.

4.3.1 Regulation

The regulation objective is to drive the membrane potential x_1 to a constant desired equilibrium value $x_{1,ref}$. Since the reference is constant, its time derivative is zero ($\dot{x}_{1,ref} = 0$) and one can define the error as:

$$e = x_{1,ref} - x_1 \quad (4.11)$$

Differentiating the error with respect to time, yields:

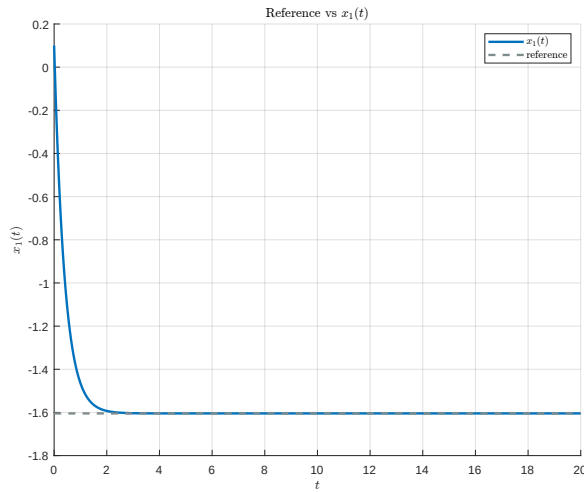
$$\dot{e} = \dot{x}_{1,ref} - \dot{x}_1 = -\dot{x}_1 \quad (4.12)$$

Recalling that the feedback linearization law results in the decoupled dynamic $\dot{y} = \dot{x}_1 = v$, the error derivative becomes:

$$\dot{e} = -v \quad (4.13)$$

To ensure exponential convergence of the error to zero, we impose the following stable dynamics:

$$\dot{e} + K_p e = 0 \implies v = K_p e \quad (4.14)$$



(a) Regulation performance

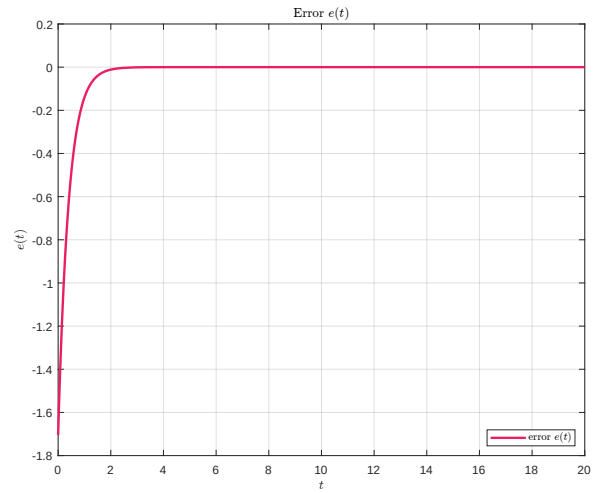
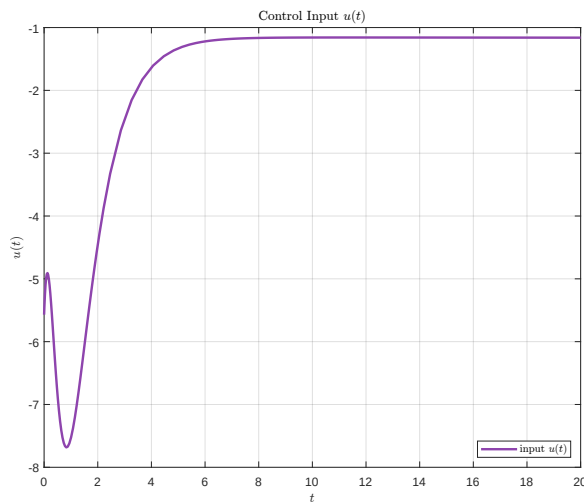
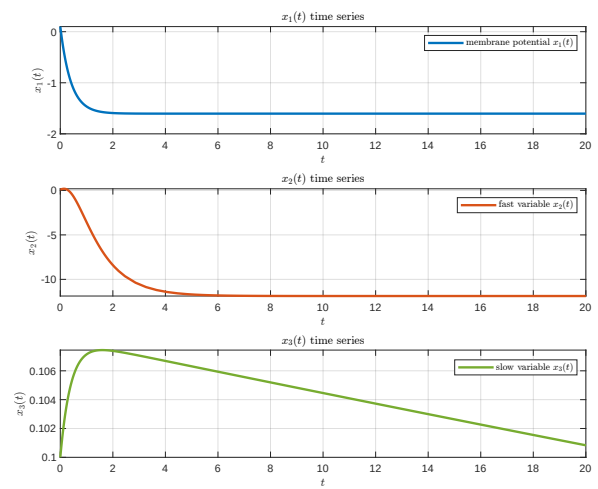
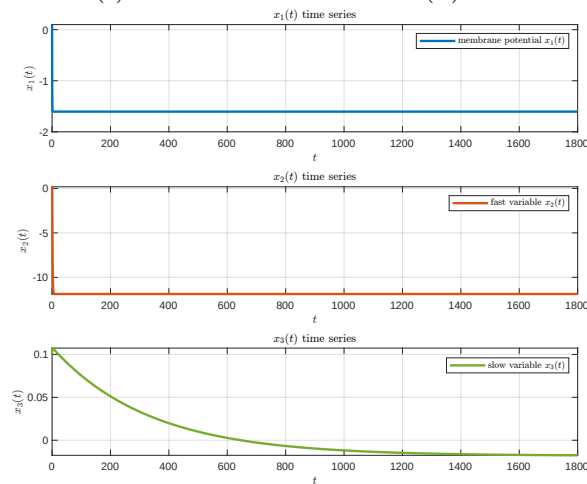
(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) State variables (x_1, x_2, x_3) (e) State variables (extended) (x_1, x_2, x_3)

Figure 4.5: FBL regulation results under nominal conditions: (a) membrane potential response to the setpoint, (b) evolution of the regulation error, (c) nonlinear control effort $u(t)$, and (d-e) complete state variable trajectories.

The resulting auxiliary control law is:

$$v = K_p(x_{1,ref} - x_1) \quad (4.15)$$

The tuning parameter $K_p > 0$ determines the speed of the response. The solution to the error equation, $e(t) = e(0)e^{-K_p t}$, implies that the settling time is inversely proportional to the gain K_p . In the Simulink implementation, K_p is adjusted to balance rapid convergence against the magnitude of the control effort u .

A purely proportional auxiliary control law ($K_I = 0$, $K_D = 0$) is intentionally selected here to evaluate the baseline performance of exact algebraic cancellation, leaving integral augmentation as a future extension for handling unmodeled dynamics.

The resulting regulation performance is illustrated in Figure 4.5, where an additional plot demonstrates the stability of the internal state variables over an extended time horizon, confirming the convergence of the zero dynamics.

The specifications are met, with $T_s = 1.54$, $S\% = 0.0\%$, with only a proportional controller ($K_I = 0$, $K_D = 0$) with $K_P = 2.5$.

4.3.2 Tracking

In the tracking problem, the control objective is to ensure the membrane potential x_1 follows a time-varying reference signal $x_{1,ref}(t)$. Define the time-varying tracking error as:

$$e(t) = x_{1,ref}(t) - x_1(t) \quad (4.16)$$

Differentiating with respect to time, one obtains:

$$\dot{e}(t) = \dot{x}_{1,ref}(t) - \dot{x}_1(t) \quad (4.17)$$

Dropping the time dependence for compactness, recalling that the feedback linearization law results in the decoupled dynamic $\dot{y} = \dot{x}_1 = v$, the error derivative becomes:

$$\dot{e} = \dot{x}_{1,ref} - v \quad (4.18)$$

To ensure exponential convergence of the error to zero, we impose the following stable dynamic:

$$\dot{e} + K_p e = 0 \implies v = K_p e + \dot{x}_{1,ref} \quad (4.19)$$

The auxiliary control input v is therefore designed as:

$$v = \dot{x}_{1,ref} + K_p(x_{1,ref} - x_1) \quad (4.20)$$

This control law incorporates a feedforward component $\dot{x}_{1,ref}$ to compensate for the reference trajectory's motion, while the proportional term K_p corrects for any instantaneous deviations. As with the regulation case, the tracking performance is globally stable due to the previously verified stability of the system's internal dynamics.

The specifications are met, with $RMSE = 0.03$, $e_{max} = 0.17$, with parameters $K_P = 2.5$, $K_I = 0$, $K_D = 0$.

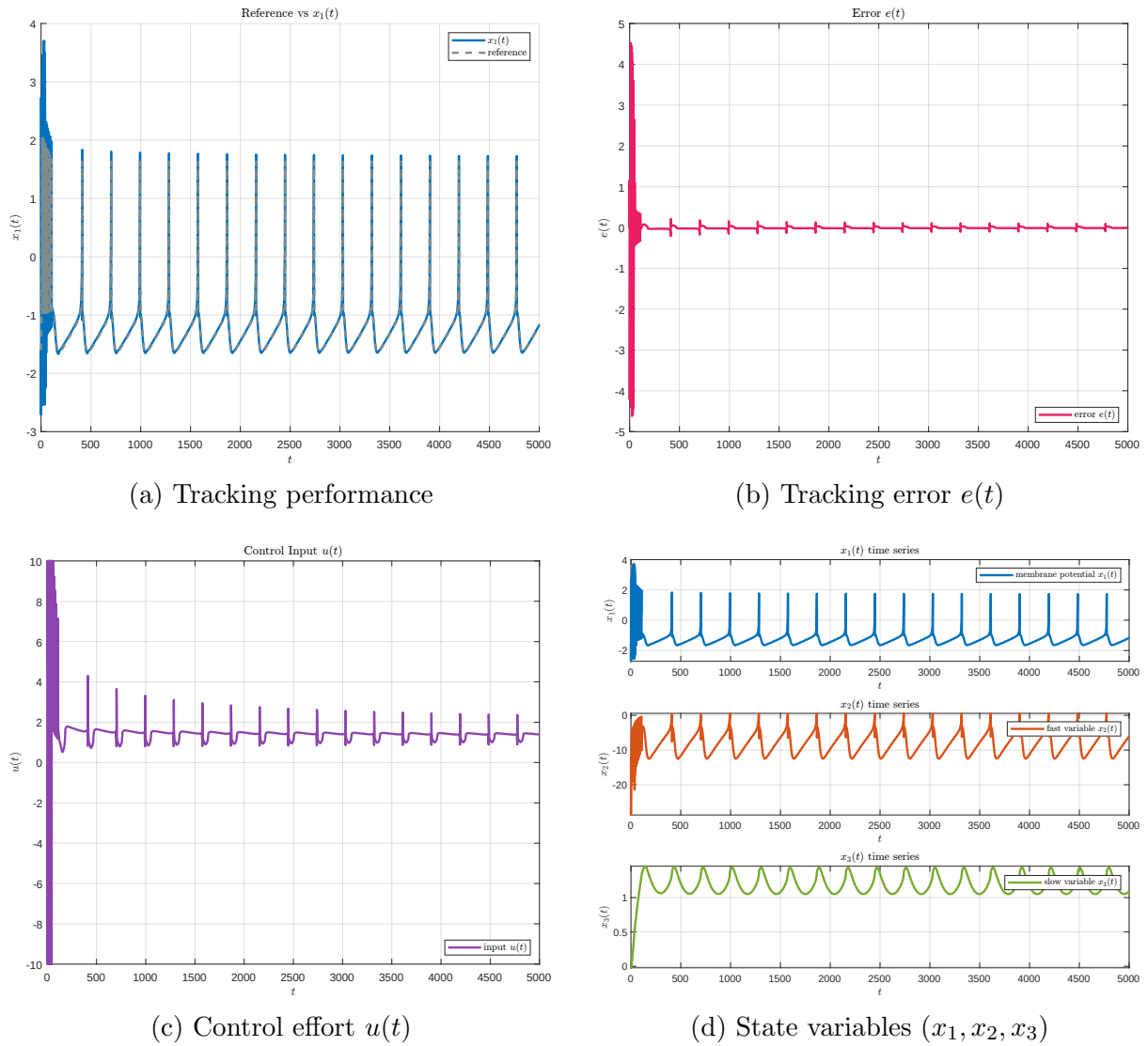


Figure 4.6: FBL tracking results under nominal conditions: (a) membrane potential response, (b) control error, (c) control effort, and (d) full state evolution.

4.4 Sliding Mode Control

To ensure robustness against unmodeled dynamics, parameter uncertainties, and external disturbances, a **Sliding Mode Control (SMC)** strategy is implemented. The implementation of the control architecture is addressed in `hr_sliding_mode_control.slx`, as depicted in Figure 4.7.

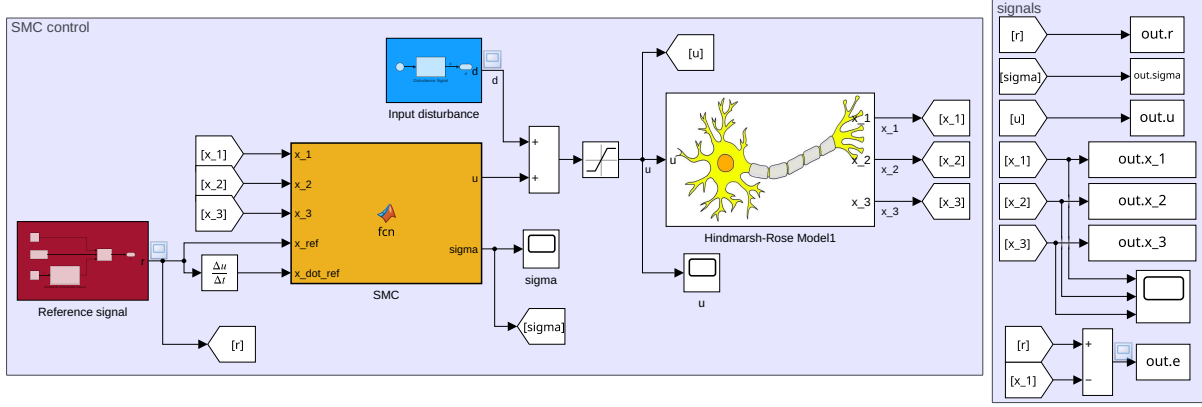


Figure 4.7: Simulink implementation of the Sliding mode Controller for the Hindmarsh-Rose model.

4.4.1 Regulation

The objective of the regulation problem is to stabilize the membrane potential x_1 at a constant reference $x_{1,ref}$.

- **Choice of the switching function:** We define the sliding surface $\sigma(\mathbf{x})$ such that the equilibrium $\sigma = 0$ implies $x = x_{1,ref}$:

$$\sigma(\mathbf{x}) = p_1(x_1 - x_{1,ref}) \quad (4.21)$$

where $p_1 > 0$.

- **Attractivity of the sliding surface:** To ensure the control input u can steer the system toward the surface, the transversality condition must hold:

$$\mathcal{L}_g\sigma = \nabla\sigma \cdot \mathbf{g}(\mathbf{x}) = \begin{bmatrix} -p_1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = p_1 \neq 0 \quad (4.22)$$

As $\mathcal{L}_g\sigma \neq 0$ for all states, the surface is globally reachable.

- **Verification of sliding dynamics:** The equivalent control u_{eq} is determined by solving $\dot{\sigma} = 0$:

$$\dot{\sigma} = \mathcal{L}_f\sigma + \mathcal{L}_g\sigma u_{eq} = 0 \implies u_{eq} = -\frac{\mathcal{L}_f\sigma}{\mathcal{L}_g\sigma} \quad (4.23)$$

Substituting the vector fields for the Hindmarsh-Rose model:

$$u_{eq} = -\frac{p_1(x_2 - ax_1^3 + bx_1^2 - x_3 + I)}{p_1} = -x_2 + ax_1^3 - bx_1^2 + x_3 - I \quad (4.24)$$

The discontinuous switching term u_{sw} is added to ensure reachability in finite time and provide robustness. To mitigate the chattering phenomenon inherent in high-frequency switching, a saturation function $\text{sat}(\cdot)$ is employed to create a boundary layer around the sliding manifold:

$$u_{sw} = -K_{smc} \text{sat}\left(\frac{\sigma}{\phi}\right) \quad (4.25)$$

where K_{smc} represents the controller gain and ϕ is the boundary layer thickness. For this implementation, the boundary layer was set to $\phi = 0.01$. The saturation function is defined as:

$$\text{sat}\left(\frac{\sigma}{\phi}\right) = \begin{cases} \frac{\sigma}{\phi} & \text{if } |\sigma| \leq \phi \\ \text{sgn}(\sigma) & \text{if } |\sigma| > \phi \end{cases} \quad (4.26)$$

The resulting control law is $u = u_{eq} + u_{sw}$.

The resulting regulation performance is illustrated in Figure 4.8, where an additional plot demonstrates the stability of the internal state variables over an extended time horizon, confirming the robustness of the sliding mode controller against persistent chattering.

4.4.2 Tracking

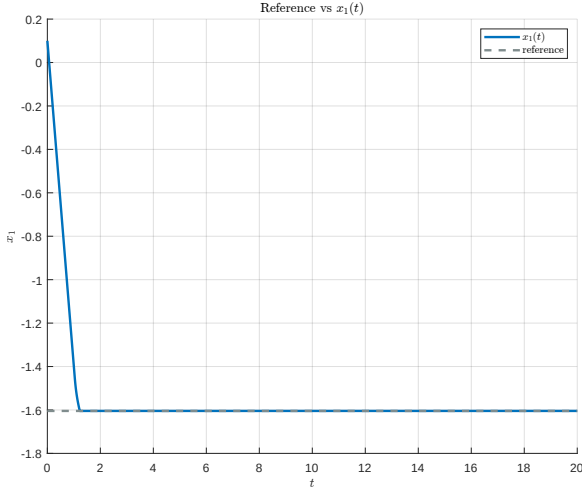
For the trajectory tracking task, the sliding surface is modified to incorporate the time-varying reference $x_{1,ref}(t)$. To achieve higher-order tracking precision, the switching function $\sigma(\mathbf{x}, t)$ is defined as:

$$\sigma(\mathbf{x}, t) = p_1(x_1 - x_{1,ref}) \quad (4.27)$$

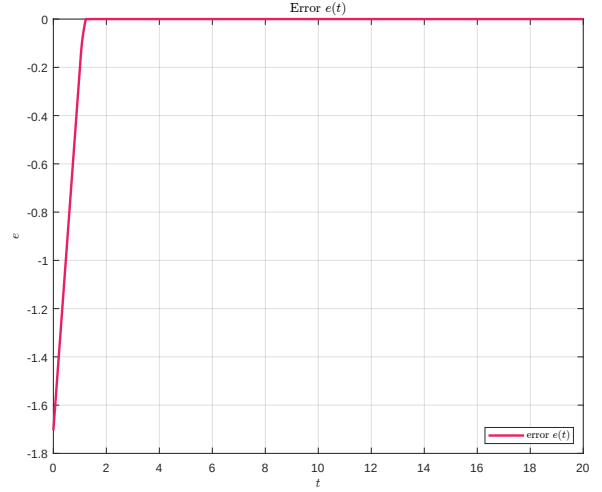
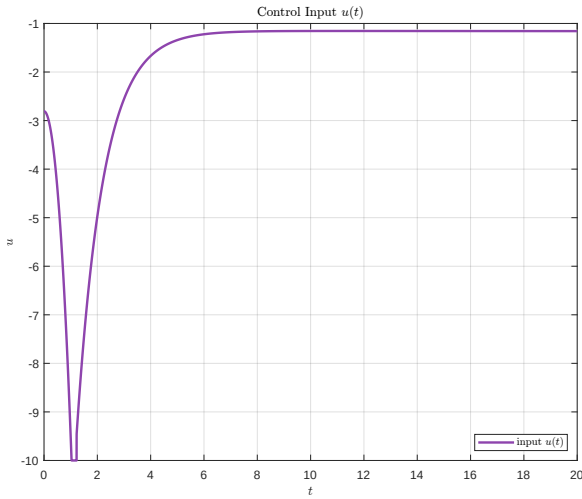
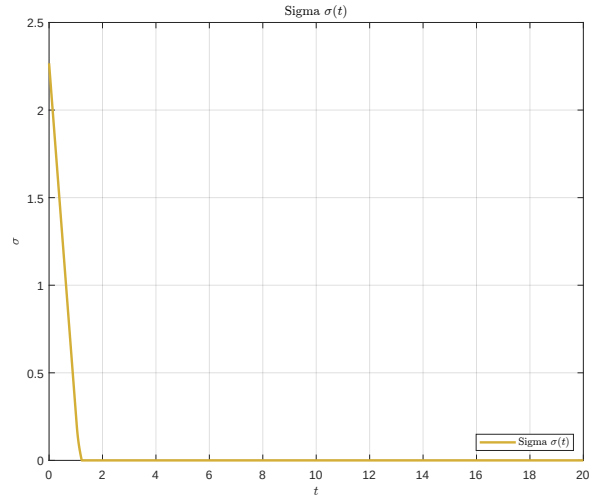
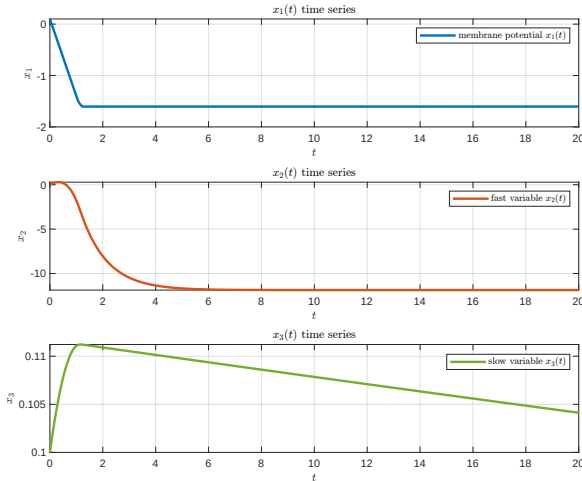
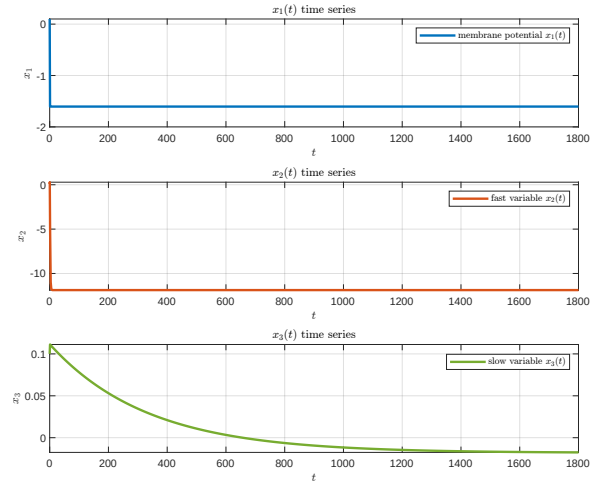
Since the relative degree of the sliding function is 1, the control input u appears directly in the first derivative $\dot{\sigma}$. For this reason, the term $p_2(\dot{x}_1 - \dot{x}_{1,ref})$ is not included. It is then possible to compute the equivalent control action, using Utkin's method as follows:

$$\dot{\sigma}(\mathbf{x}, t) = 0 \iff u_{eq} = -x_2 + ax_1^3 - bx_1^2 + x_3 - I + \dot{x}_{1,ref} \quad (4.28)$$

The control synthesis follows the same logic as the regulation case, with u_{eq} now including the feedforward terms of the reference derivatives. Finally, the stability of the internal dynamics is preserved under the bounded reference trajectory $x_{1,ref}(t)$, as the subsystem remains stable.



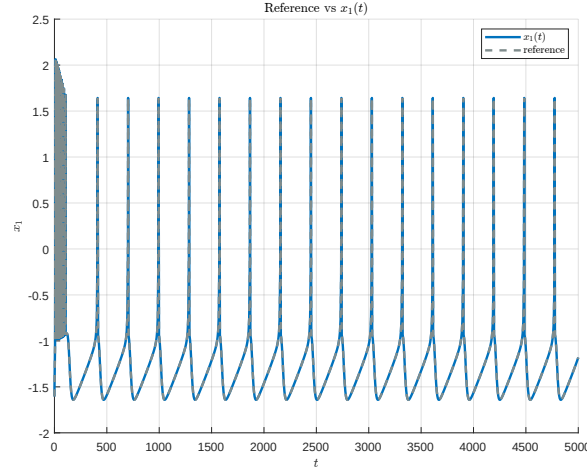
(a) Regulation performance

(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) Sliding surface $\sigma(t)$ (e) State variables (x_1, x_2, x_3) 

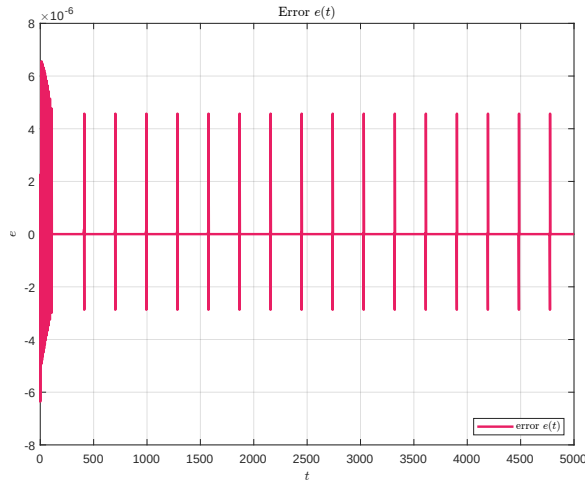
(f) Extended state evolution

Figure 4.8: SMC regulation results under nominal conditions: (a) membrane potential, (b) regulation error, (c) robust control effort, (d) evolution of the switching function, and (e-f) state trajectories.

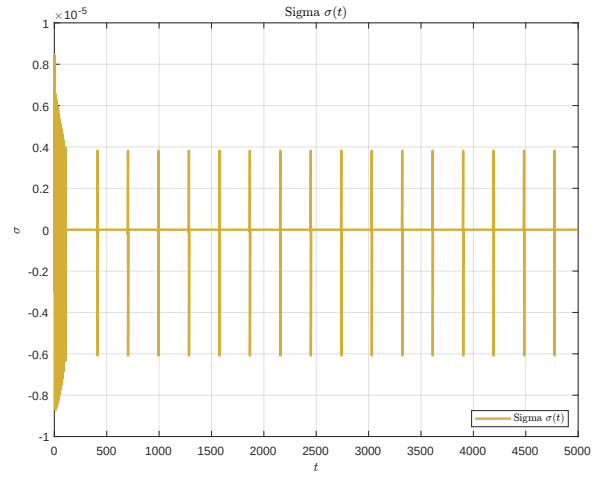
The specifications are met, with $T_s = 1.16$, $S_{\%} = 0.0\%$, using $K_{smc} = 1.5$, $p_1 = 1.33$.



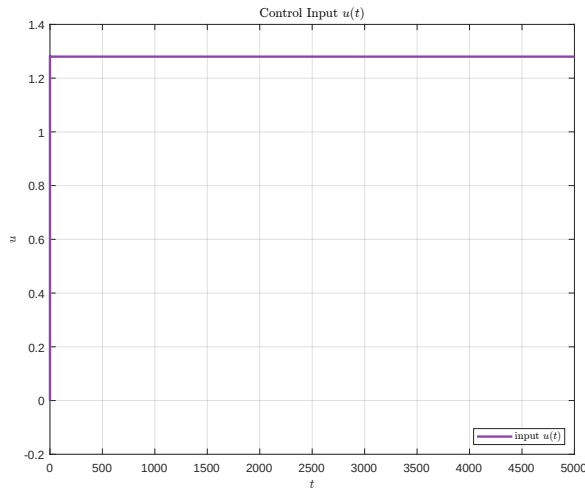
(a) Tracking performance: Reference signal $x_{1,ref}(t)$ vs. membrane potential $x_1(t)$.



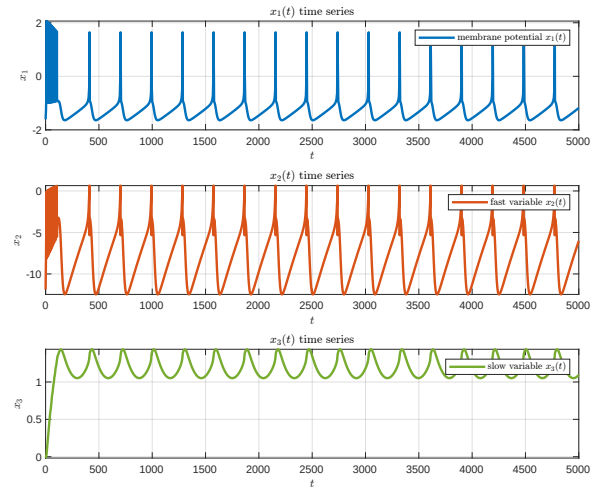
(b) Tracking error $e(t)$



(c) Sliding surface $\sigma(t)$



(d) Control effort $u(t)$



(e) State variables (x_1, x_2, x_3)

Figure 4.9: SMC tracking performance: tracking behavior (a) is supported by the error dynamics (b), surface reachability (c), control signal (d), and internal state stability (e).

The specifications are met, with $RMSE = 0.00$, $e_{max} = 4 \times 10^{-6}$, $K_{smc} = 1.5$, $p_1 = 1.33$.

Chapter 5

Robustness Analysis

While the control strategies designed in the previous chapter demonstrate high performance under nominal conditions, practical neurostimulation applications must contend with environmental noise and biological variability. This chapter evaluates the robustness of the **PID**, **Feedback Linearization (FBL)**, and **Sliding Mode Control (SMC)** strategies against external input disturbances and model uncertainties.

Three distinct test scenarios were implemented across both regulation and tracking tasks:

- **Input Disturbance Rejection:** An external sinusoidal current $d(t) = A \sin(\omega t)$ was injected into the control input channel to simulate environmental noise or non-ideal electrode behavior. For these simulations, the disturbance was defined with an amplitude $A = 0.5$ and an angular frequency $\omega = 2\pi(0.2)$ rad/s.
- **Parametric Uncertainties:** To account for the inherent variability in neuronal physical properties, a 10% uncertainty was introduced into the system parameters (a, b, c, d) . The controllers were designed using nominal values, while the actual plant dynamics operated with biased coefficients.
- **Combined Uncertainties in Tracking:** For the trajectory tracking task, the synchronization fidelity was tested under the simultaneous influence of parametric mismatches and input disturbances, assessing the controllers' ability to maintain a pinning-control regime despite significant model degradation.

The performance of each strategy was optimized through iterative parameter tuning to balance disturbance rejection against control effort, with specific gains and metrics reported in the subsequent sections.

5.1 PID control

5.1.1 Regulation with input disturbance

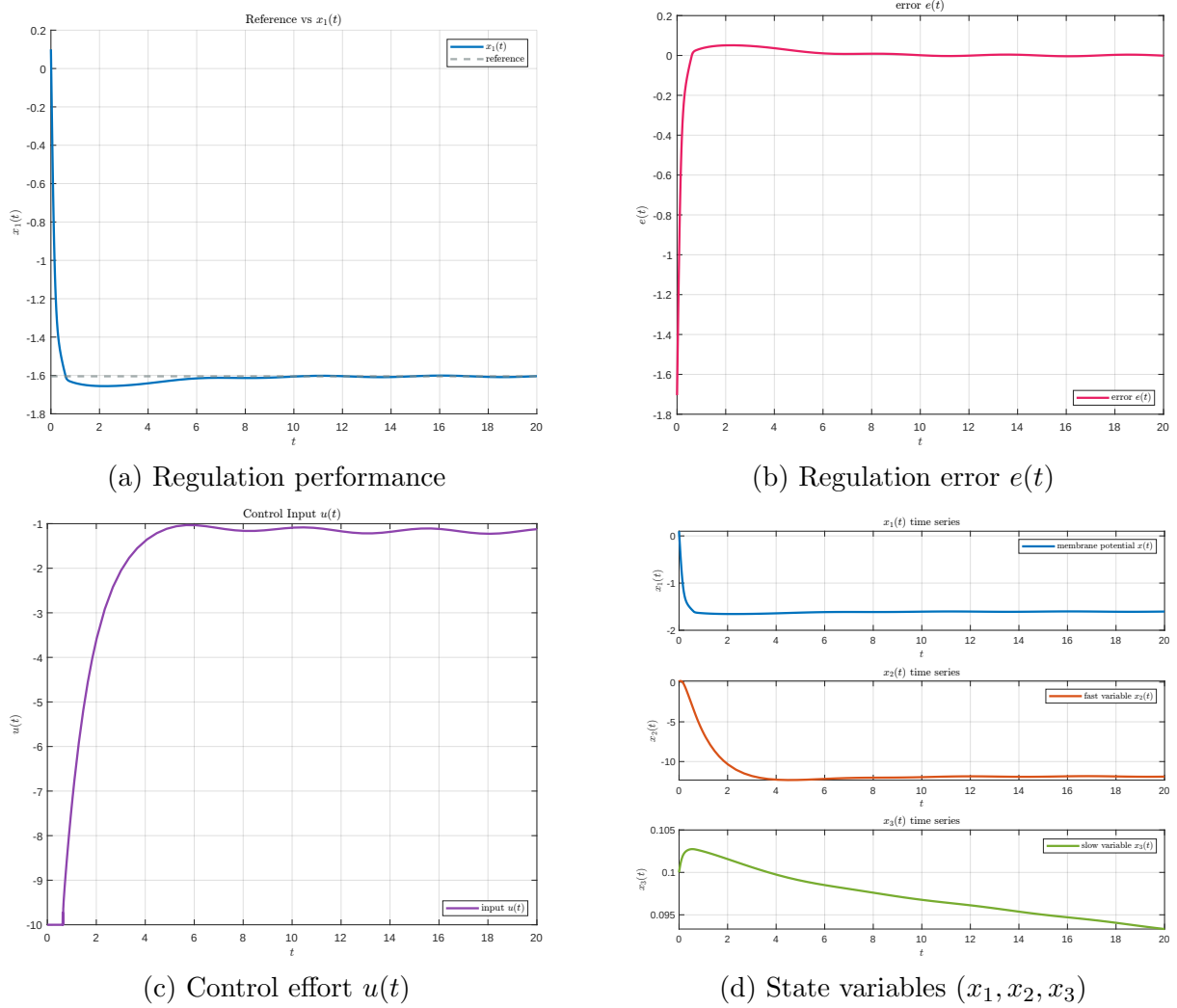
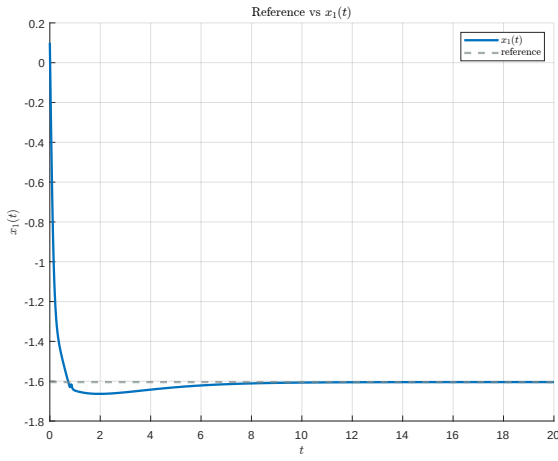


Figure 5.1: PID regulation results under sinusoidal input disturbance: (a) regulation response, (b) control error, (c) control action, and (d) state evolution.

Parameter	Value
K_p	108.97
K_i	49.48
K_d	2.67
Metric	Result
T_s	4.07
S_p (%)	3.04%

Table 5.1: PID tuning parameters and regulation performance under input disturbance.

5.1.2 Regulation with parametric uncertainties



(a) Regulation performance

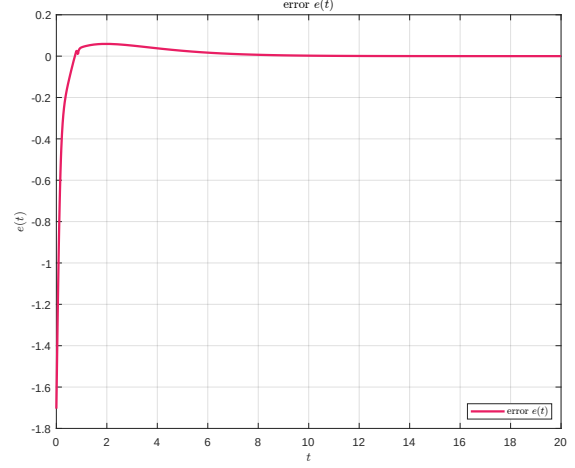
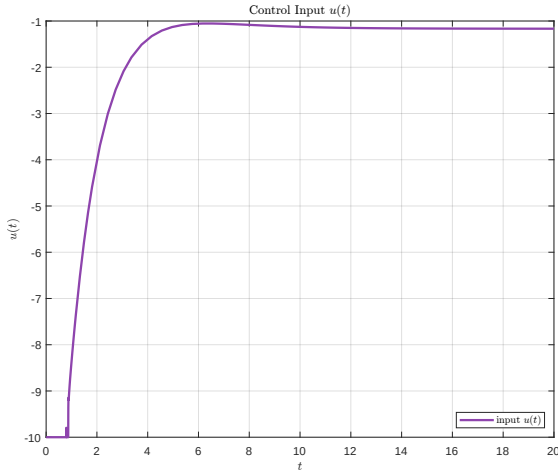
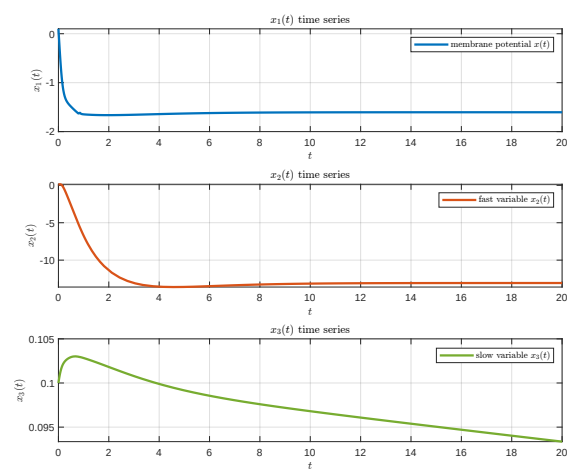
(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) State variables (x_1, x_2, x_3)

Figure 5.2: PID regulation results under 10% parametric uncertainties.

Parameter	Value
K_p	108.97
K_i	49.48
K_d	2.67
Metric	Value
T_s	4.16
S_p (%)	3.46%

Table 5.2: PID performance metrics under 10% parametric uncertainty.

5.2 Feedback Linearization

5.2.1 Regulation with input disturbance

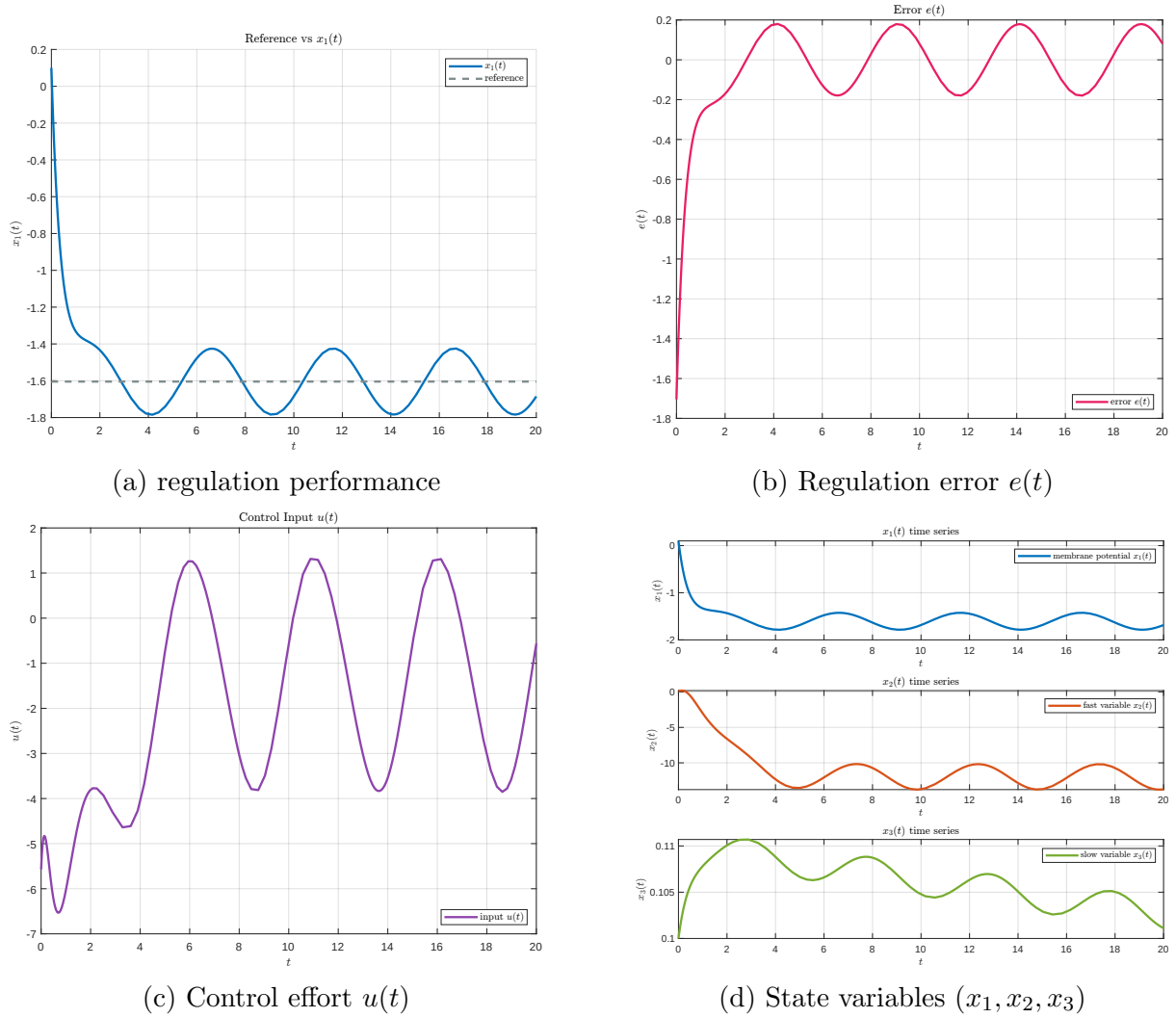
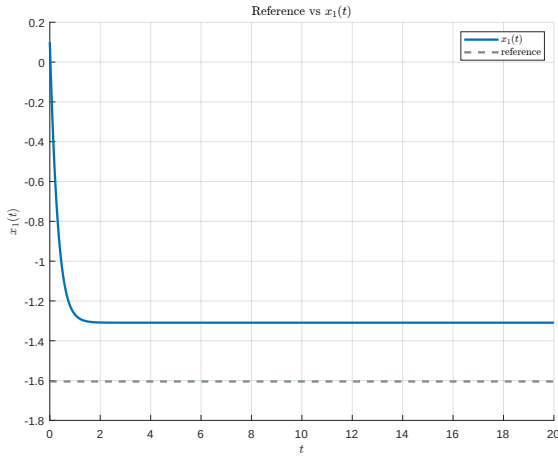


Figure 5.3: FBL regulation results under sinusoidal input disturbance.

Metric	Value
K_p (Auxiliary)	2.5
T_s	20.0
S_p (%)	5.58%

Table 5.3: FBL regulation metrics under sinusoidal input disturbance.

5.2.2 Regulation with parametric uncertainties



(a) Regulation performance

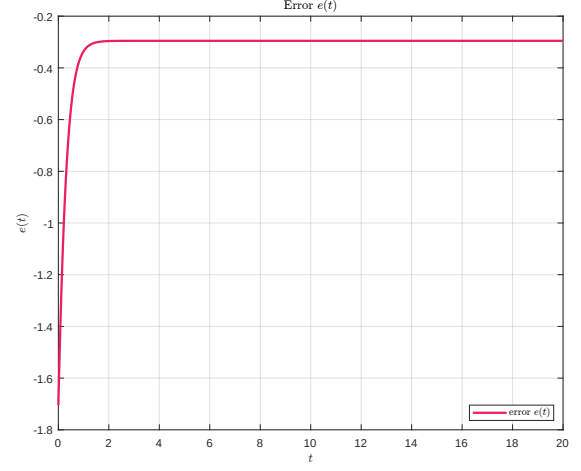
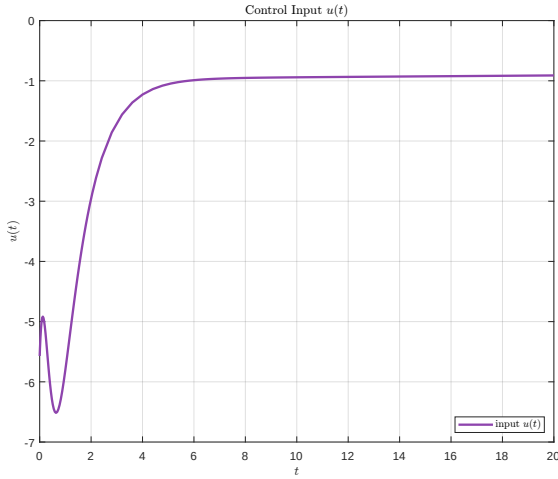
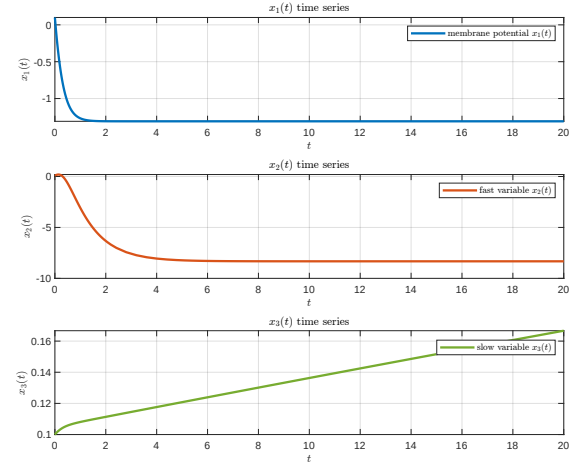
(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) State variables (x_1, x_2, x_3)

Figure 5.4: FBL regulation under 10% parametric uncertainties: model mismatch results in a slight steady-state offset due to imperfect cancellation of nonlinearities.

Metric	Value
K_p (Auxiliary)	2.5
T_s	20.0
S_p (%)	0.0%

Table 5.4: FBL regulation metrics under parameters uncertainties.

5.2.3 Tracking with input disturbance

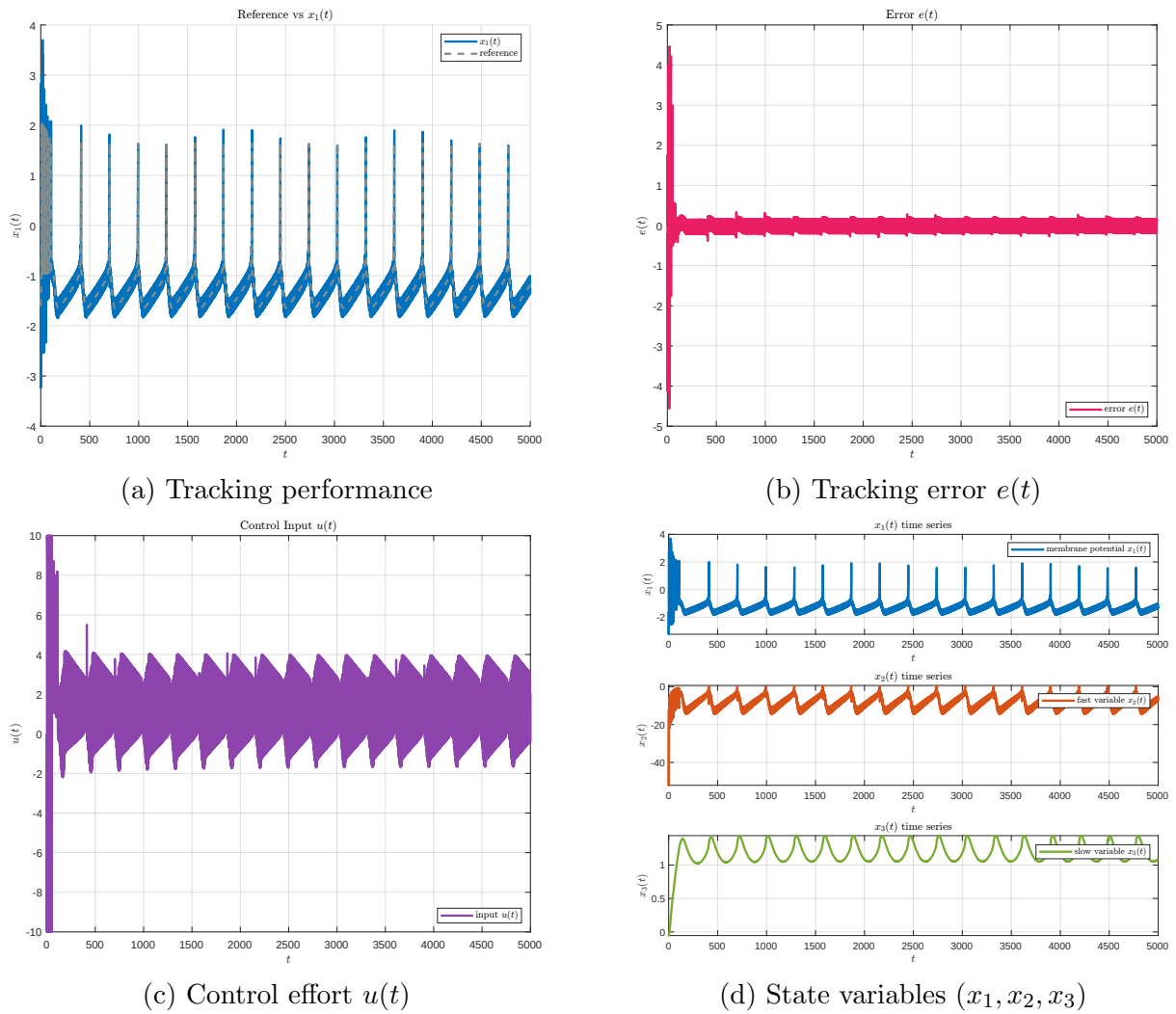


Figure 5.5: FBL trajectory tracking under sinusoidal input disturbance: synchronization fidelity is challenged by the exogenous noise during firing phases.

Metric	Value
RMSE	0.13
e_{max}	0.30

Table 5.5: FBL tracking metrics under sinusoidal input disturbance.

5.2.4 Tracking with parametric uncertainties

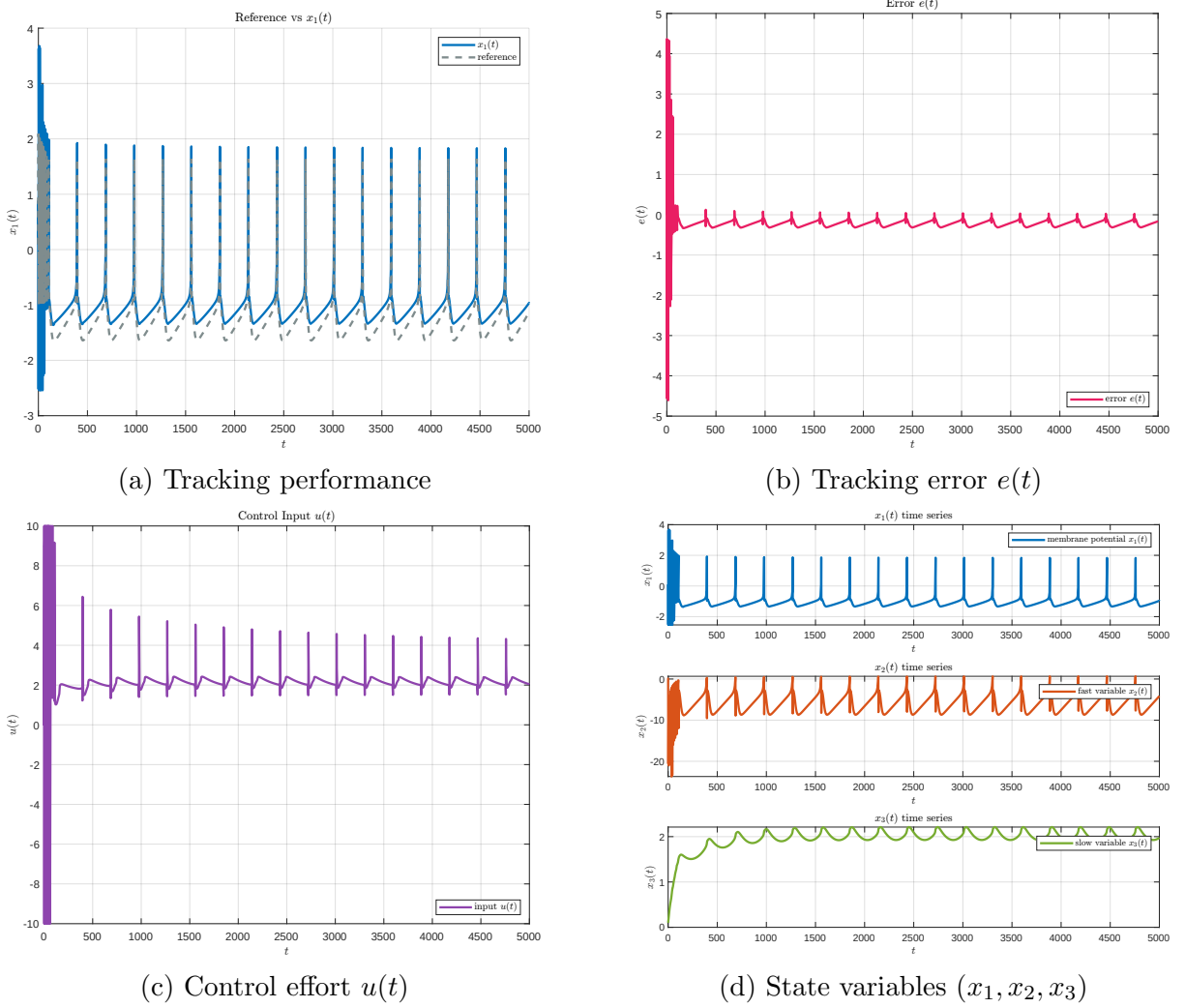


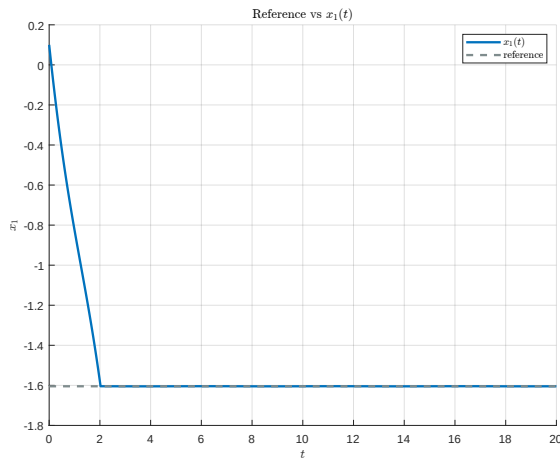
Figure 5.6: FBL trajectory tracking under 10% parametric uncertainties: the degradation of synchronization performance is visible during the spiking transitions.

Metric	Value
RMSE	0.21
e_{max}	0.32

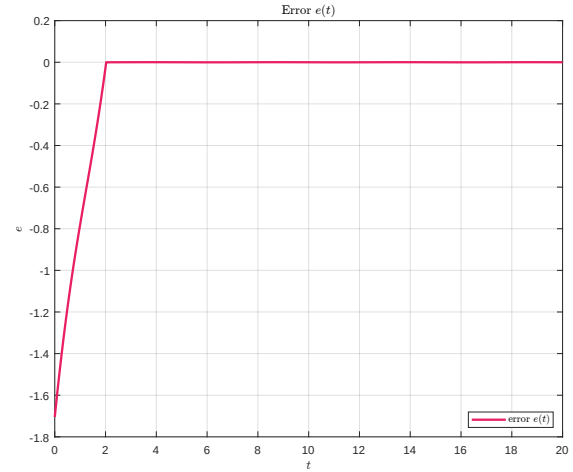
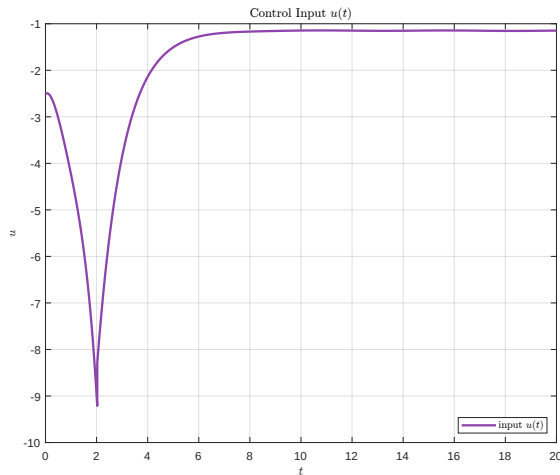
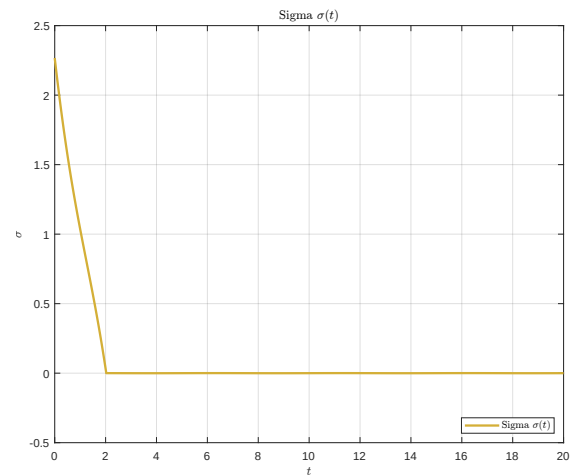
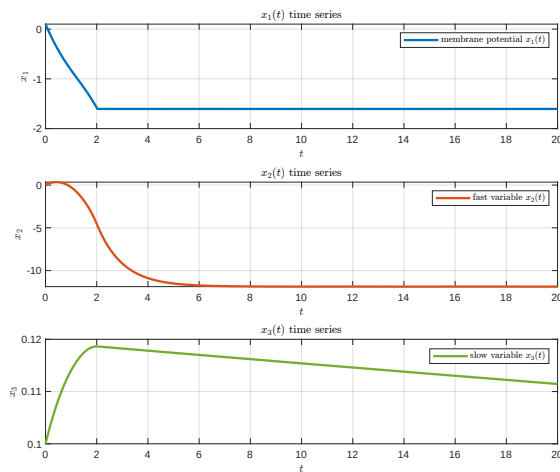
Table 5.6: FBL tracking metrics under 10% parametric uncertainty.

5.3 Sliding Mode Control

5.3.1 Regulation with input disturbance



(a) Regulation performance

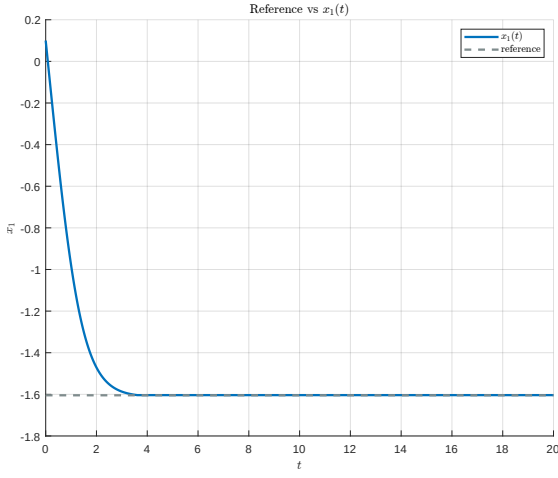
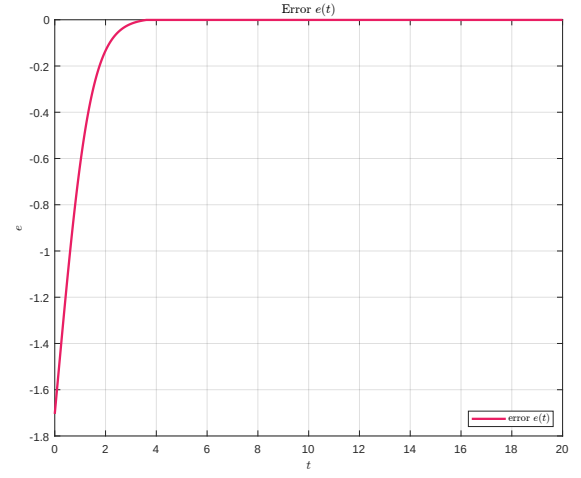
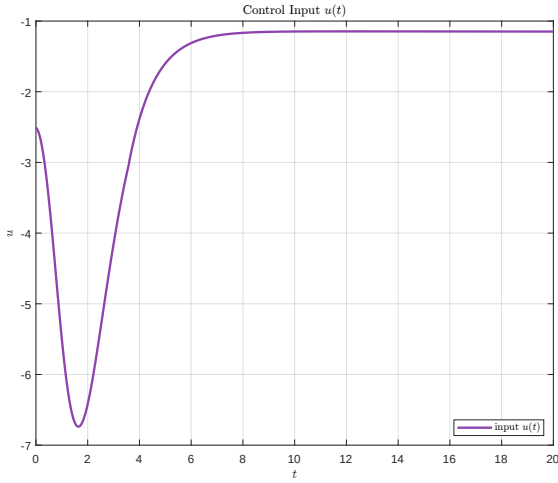
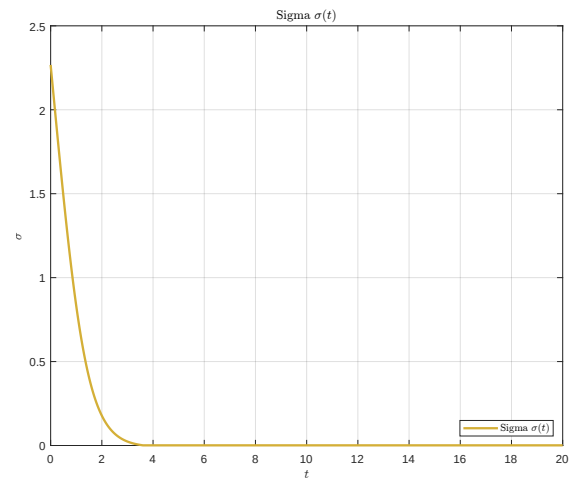
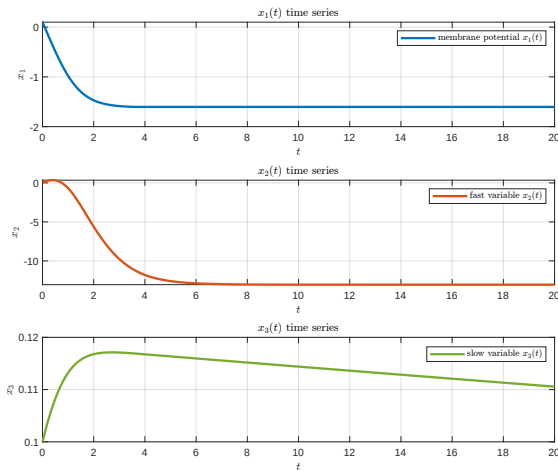
(b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) Sliding surface $\sigma(t)$ (e) State variables (x_1, x_2, x_3)

Parameter/Metric	Value
K_{smc}	1.2
p_1	1.33
ϕ	0.01
T_s	1.44
S_p (%)	0.01%

(f) Numerical performance

Figure 5.7: SMC regulation results under sinusoidal input disturbance: (a-b) setpoint maintenance, (c-d) control effort and sliding surface reachability, (e) state evolution, and (f) performance summary.

5.3.2 Regulation with parametric uncertainties

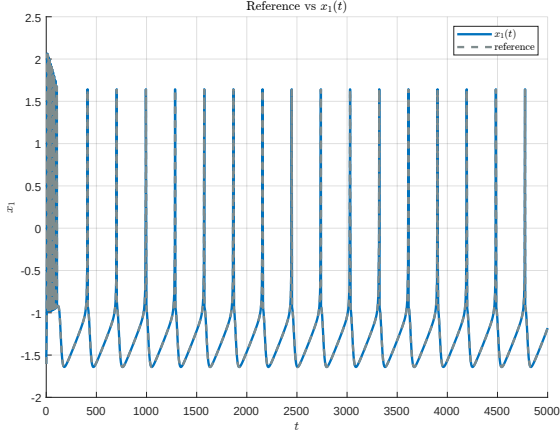
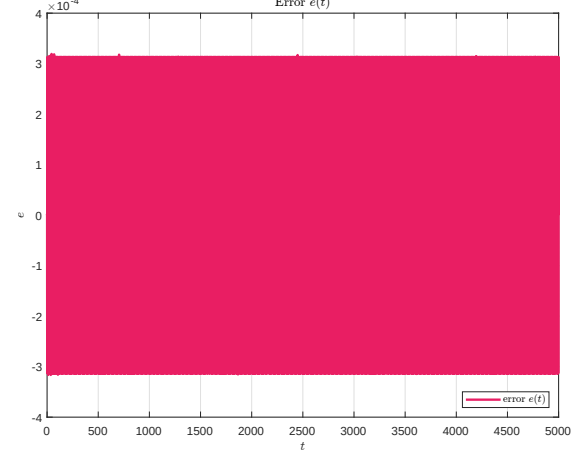
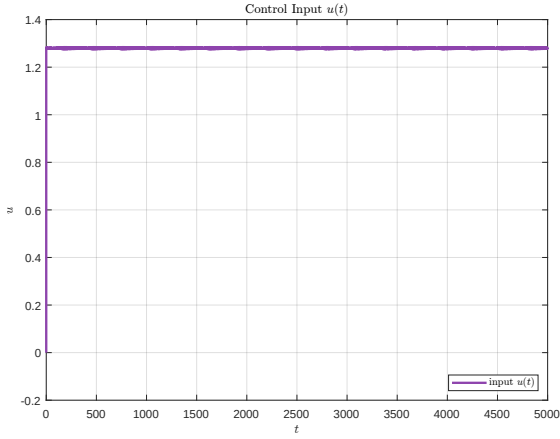
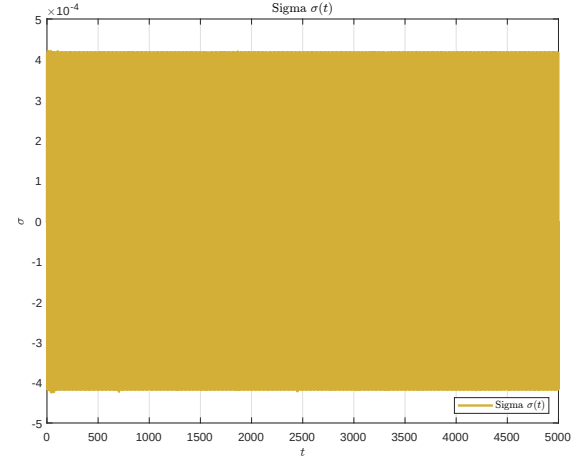
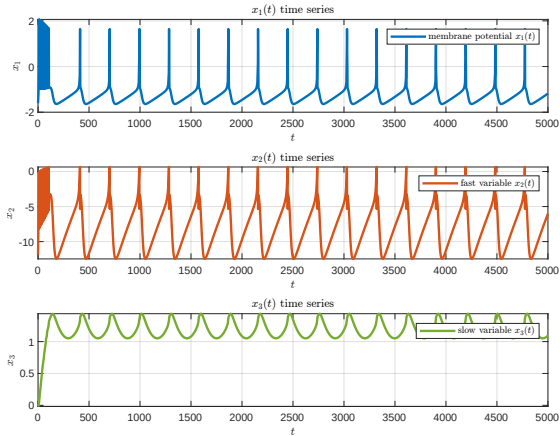
(a) Response of membrane potential $x_1(t)$ (b) Regulation error $e(t)$ (c) Control effort $u(t)$ (d) Sliding surface $\sigma(t)$ (e) State variables (x_1, x_2, x_3)

Parameter/Metric	Value
K_{smc}	1.2
p_1	1.33
ϕ	0.01
T_s	2.71
S_p (%)	0.00%

(f) Numerical performance

Figure 5.8: SMC regulation under 10% parametric uncertainties: (a-b) setpoint maintenance, (c-d) control effort and surface invariance, (e) state evolution, and (f) performance summary.

5.3.3 Tracking with input disturbance

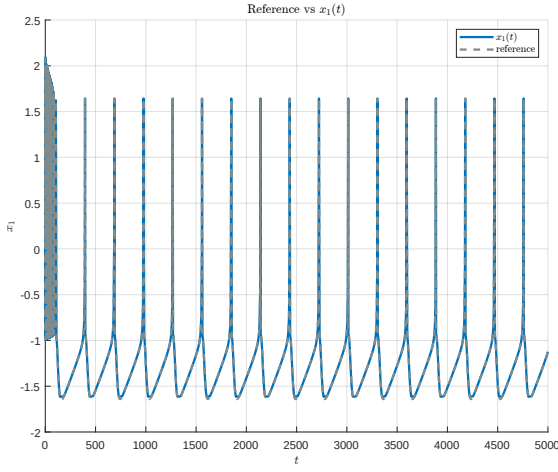
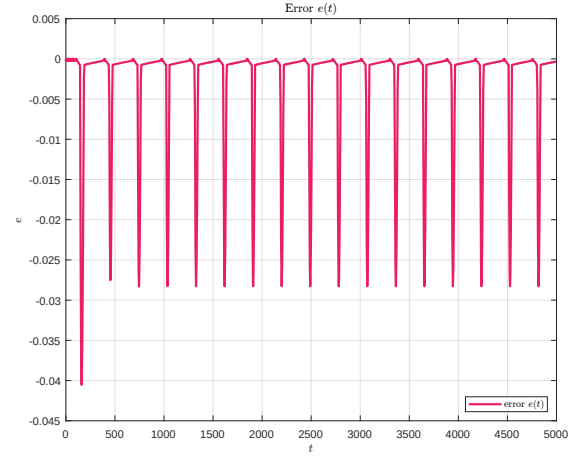
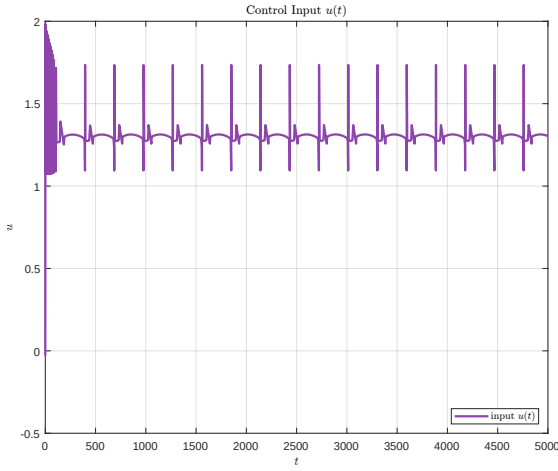
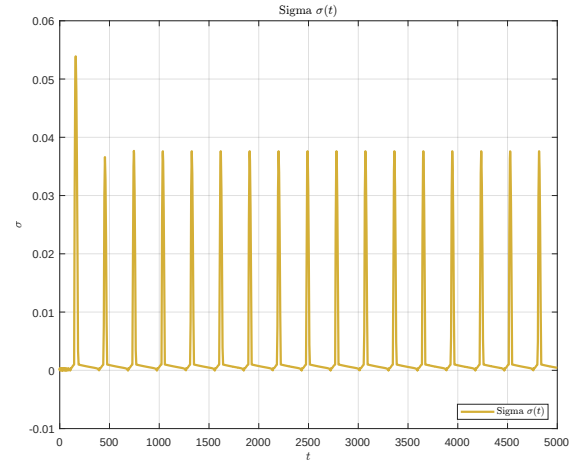
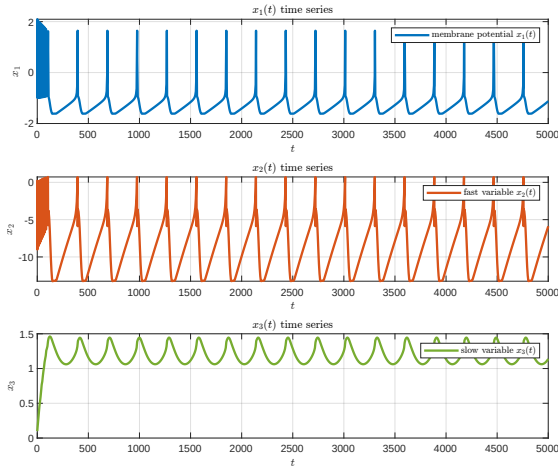
(a) Tracking performance $x_1(t)$ (b) Tracking error $e(t)$ (c) Control effort $u(t)$ (d) Sliding surface $\sigma(t)$ (e) State variables (x_1, x_2, x_3)

Metric	SMC Result
K_{smc}	1.2
p_1	1.33
ϕ	0.01
RMSE	1.7×10^{-4}
e_{max}	2.5×10^{-4}

(f) Numerical performance

Figure 5.9: SMC tracking performance under sinusoidal input disturbance: (a-b) synchronization fidelity and error, (c-d) robust control action and manifold reachability, (e) state evolution, and (f) tracking metrics.

5.3.4 Tracking with parametric uncertainties

(a) Tracking performance $x_1(t)$ (b) Tracking error $e(t)$ (c) Control effort $u(t)$ (d) Sliding surface $\sigma(t)$ (e) State variables (x_1, x_2, x_3)

Metric	SMC Result
K_{smc}	1.2
p_1	1.33
ϕ	0.01
RMSE	4×10^{-4}
e_{max}	6×10^{-4}

(f) Numerical performance

Figure 5.10: SMC tracking performance under 10% parametric uncertainties: (a-f) comprehensive analysis of robust synchronization despite model degradation.

Chapter 6

Performance Comparison

This chapter presents a comprehensive quantitative comparison of the implemented control strategies: **Proportional-Integral-Derivative (PID)**, **Input-Output Feedback Linearization (FBL)**, and **Sliding Mode Control (SMC)**. The evaluation is conducted across regulation and tracking tasks, assessing how each controller handles nominal conditions, external disturbances, and parametric uncertainties to objectively identify the most suitable candidate for neurostimulation tasks.

6.1 Performance Metrics

To objectively assess and compare the performance of each strategy, the following metrics were recorded during the simulations:

- **Settling Time (T_s)**: The time required for the tracking error to stay within a 2% boundary of the step magnitude.
- **Percentage Overshoot (S_p)**: The maximum peak value of the response relative to its steady-state value.
- **Steady-State Error (e_∞)**: The average absolute error measured during the final second of the regulation simulation.
- **Peak Control Effort (u_{max})**: The maximum absolute value of the control input current required.
- **Control Energy (J_u)**: The integral of the squared control input ($\int_0^T u^2(t)dt$), representing the total energy consumption.
- **RMSE**: Root Mean Square Error, used in tracking tasks to evaluate the average deviation from the reference trajectory.
- **Maximum Error (e_{max})**: The highest absolute error recorded during the tracking phase.

6.2 Regulation Performance

This section evaluates the controllers under nominal conditions, input disturbances ($A = 0.5$ sinusoidal), and 10% parametric uncertainties.

Controller	T_s (t.u.)	S_p (%)	e_∞	u_{max} (c.u.)	J_u
<i>Nominal Conditions</i>					
PID	3.59	3.05	2.2×10^{-6}	10.00	149.76
FBL	1.54	0.00	$< 10^{-15}$	7.68	119.63
SMC	1.16	0.00	$< 10^{-15}$	10.00	129.49
<i>Sinusoidal Input Disturbance</i>					
PID	4.07	3.04	8.10×10^{-4}	10.0	149.72
FBL	20.00	5.58	8.04×10^{-2}	10.0	172.91
SMC	1.44	0.01	1.6×10^{-7}	10.00	123.75
<i>Parametric Uncertainties</i>					
PID	4.16	3.46	2.2×10^{-6}	10.00	166.46
FBL	20.00	0.00	0.29	6.51	75.61
SMC	2.71	0.00	7.4×10^{-4}	6.73	119.91

Table 6.1: Quantitative comparison of control strategies for the regulation task.

The quantitative comparison presented in the regulation table reveals the performance trade-offs between the three control architectures when subjected to nominal conditions and external disturbances or uncertainties. Moreover remember that the time units and current are dimensionless (t.u)

- **Nominal Performance and Efficiency:** Under ideal conditions, the **FBL** controller proves to be the most energy-efficient, achieving a settling time of 1.54 with the lowest control energy ($J_u = 119.63$) and a peak effort well below the 10 c.u. limit. While **SMC** is the fastest ($T_s = 1.16$ t.u.), its aggressive nature is reflected in a higher energy consumption ($J_u = 129.49$) and the immediate saturation of the control input at 10 c.u during the reaching phase.
- **Response to Sinusoidal Disturbance:** The introduction of an external current oscillation highlights a major vulnerability in the **FBL** strategy. Because it relies on precise algebraic cancellation, the disturbance causes the settling time to jump to 20.00 t.u. with a significant steady-state error (8.04×10^{-2}). In contrast, the **SMC** maintains superior precision ($e_\infty \approx 10^{-7}$) and a fast settling time (1.44 t.u.), demonstrating the "invariance property" where the robust switching term successfully rejects input perturbations.
- **Impact of Parametric Uncertainties:** When a 10% bias is applied to the system parameters, the **PID** controller shows its classic advantage; its integral action keeps the steady-state error at 10^{-6} , whereas the standard **FBL** (lacking an integral term) fails significantly with an error of 0.29. The **SMC** manages to remain stable and precise ($e_\infty = 7.4 \times 10^{-4}$), proving it to be the most balanced architecture for biological systems where exact modeling of neuron parameters is often impossible.
- **Control Effort and Saturation:** Both **PID** and **SMC** consistently reach the 10 c.u. saturation limit during the initial transient phase to minimize settling time. **FBL** is the only controller that remains within the linear region of the actuator (7.68 c.u.) under nominal conditions, suggesting it is the safest option for neurostimulation provided that the model parameters are accurately identified.

6.3 Tracking Performance

The tracking fidelity was evaluated under non-ideal conditions, including simultaneous input disturbances and parametric mismatches. The tracking reference consists of a complex neuronal firing pattern. The PID controller was not evaluated for tracking tasks, as its linear design is not suited for following a complex nonlinear reference signal.

Scenario	Metric	FBL	SMC
<i>Nominal Conditions</i>	RMSE	0.03	1.4×10^{-7}
	e_{max}	0.17	4×10^{-6}
<i>Input Disturbance</i>	RMSE	0.13	1.7×10^{-4}
	e_{max}	0.30	2.5×10^{-4}
<i>Parametric Unc.</i>	RMSE	0.21	4×10^{-4}
	e_{max}	0.32	6.0×10^{-4}

Table 6.2: Quantitative comparison of tracking performance under robustness stressors.

The quantitative results for the trajectory tracking task, summarized in Table 6.2, highlight the performance gap between the two nonlinear strategies when maintaining synchronization under stress.

- Nominal Precision and "Pinning" Capability:** Under ideal conditions, the **SMC** demonstrates near-perfect synchronization, achieving an RMSE of 1.4×10^{-7} and a peak error (e_{max}) of 4×10^{-6} . This level of precision is several orders of magnitude superior to the **FBL**, which exhibits an RMSE of 0.03. These results confirm the effectiveness of the sliding manifold in "pinning" the follower neuron to the leader's trajectory.
- Robustness to Input Disturbance:** The introduction of external current oscillations reveals a significant degradation in **FBL** performance, as its RMSE increases from 0.03 to 0.13. In contrast, the **SMC** maintains a highly accurate track with an RMSE of 1.7×10^{-4} . This behavior illustrates the *invariance property* of SMC, where the robust switching term successfully rejects external perturbations that would otherwise disrupt the synchronization.
- Sensitivity to Parametric Uncertainties:** The **FBL** strategy is highly sensitive to the 10% parametric bias, which results in its highest recorded tracking error ($RMSE = 0.21$, $e_{max} = 0.32$). This confirms that algebraic cancellation is fragile when the internal parameters of the follower neuron do not perfectly match the leader. The **SMC** remains remarkably robust in this scenario, keeping the peak error at a negligible 6.0×10^{-4} , demonstrating its superiority for biological applications where exact modeling is difficult.
- Overall Synchronization Verdict:** While **FBL** provides a smoother control action, the data indicates that **SMC** is the only controller capable of maintaining high-fidelity synchronization across all tested stressors. The ability of SMC to keep the RMSE and e_{max} consistently low, regardless of disturbance or model mismatch, makes it the optimal choice for robust neurostimulation.

Chapter 7

Conclusion

This work presented a comprehensive study of the Hindmarsh-Rose (HR) neuronal model, covering its mathematical derivation, open-loop dynamical analysis, and the synthesis and robustness evaluation of three distinct control strategies aimed at neurostimulation tasks.

The open-loop analysis established a rigorous characterization of the model's rich dynamical repertoire as a function of the external bias current I . Through equilibrium analysis, Jacobian linearization, and numerical continuation via `matcont`, two critical bifurcations were identified: a Limit Point of Cycles (LPC) at $I \approx 1.276$, giving rise to a bistable regime in which both a stable focus and a stable limit cycle coexist, and a subcritical Hopf bifurcation at $I \approx 1.306$, beyond which the resting equilibrium loses stability and all trajectories are captured by the global bursting attractor. The subsequent bifurcation analysis, conducted through Inter-Spike Interval diagrams and Poincaré sections, uncovered a period-adding cascade in the range $1.28 \leq I \leq 3.10$, followed by a chaotic window near $I \approx 3.31$ characterised by a Boundary Crisis, and finally a reverse period-doubling cascade leading to tonic spiking for $I > 3.35$.

Three control architectures were designed and validated in Simulink for both setpoint regulation and trajectory tracking tasks. The PID controller, tuned via linearisation of the HR system around the desired equilibrium, provided a reliable performance baseline, achieving a settling time of $T_s = 3.59$ with a modest overshoot of 3.05% and robustness to both sinusoidal disturbances and parametric uncertainties thanks to its integral action. The Input-Output Feedback Linearization (FBL) controller exploited the relative degree-1 structure of the system to exactly cancel the nonlinear dynamics, yielding the fastest nominal regulation ($T_s = 1.54$, zero overshoot) with the lowest energy consumption ($J_u = 119.63$). However, this exact-cancellation approach proved sensitive to both additive disturbances and model mismatch, resulting in significant steady-state errors under perturbed conditions. The Sliding Mode Controller (SMC) demonstrated the best overall robustness profile: it achieved the fastest regulation settling time ($T_s = 1.16$), near-perfect trajectory tracking under nominal conditions ($\text{RMSE} = 1.4 \times 10^{-7}$), and maintained high synchronisation fidelity even in the presence of sinusoidal disturbances and 10% parametric uncertainties, by virtue of its invariance property.

In summary, the comparative analysis confirms that no single controller is universally optimal across all criteria. FBL is the preferred choice when an accurate model is available and energy efficiency is paramount. PID offers simplicity and robustness to parameter drift via integral action. SMC represents the most reliable architecture for neurostimulation applications where biological variability and environmental noise are unavoidable, making it the recommended strategy for the closed-loop control of neuronal

dynamics in realistic settings.

Future Developments

Several directions could naturally extend the results presented in this thesis.

Trajectory tracking in different dynamical regimes. The tracking problem in this work was formulated with the leader neuron operating in the periodic spiking regime. A natural extension is to study synchronisation when the leader neuron exhibits richer dynamics, in particular chaotic bursting (e.g., at $I \approx 3.31$). In the chaotic regime, the reference signal is aperiodic and exponentially sensitive to initial conditions, which places substantially greater demands on the controller’s bandwidth and robustness. Investigating whether SMC can maintain synchronisation of a follower to a chaotic leader, and at what energy cost, would yield important insights for applications such as epileptic seizure suppression, where pathological neural activity is often chaotic in nature.

Integral augmentation of the Feedback Linearization controller. The primary weakness of the FBL strategy is the appearance of a non-negligible steady-state error under parametric uncertainties, caused by imperfect algebraic cancellation of the nonlinear terms. This limitation can be addressed by augmenting the auxiliary linear controller with an integral action, transforming it from a proportional (P) to a PI or PID structure in the linearised coordinates. The inclusion of an integral term would eliminate the offset introduced by model mismatch, recovering zero steady-state error without sacrificing the favourable energy profile of the FBL approach. A careful stability analysis of the augmented zero dynamics would be required to ensure that integral windup does not destabilise the internal states.

Pinning control for network synchronisation. The tracking problem studied here can be interpreted as a pinning control problem on a minimal two-neuron network, in which a controlled follower is driven to synchronise with an uncontrolled leader. A compelling extension is to scale this framework to a network of N HR neurons, where only a fraction of nodes (the *pinned* nodes) are equipped with a controller, while the remaining neurons are coupled passively through synaptic diffusion. The key open questions concern the minimum number and optimal placement of pinned nodes required to achieve global network synchronisation, and the effect of network topology (e.g., all-to-all, small-world, scale-free) on convergence speed and control energy. Such a framework is directly relevant to the design of Deep Brain Stimulation (DBS) protocols, where only a limited number of electrodes can be implanted and the goal is to entrain the entire target nucleus to a desired firing pattern.

These directions collectively point toward a unified framework for model-based closed-loop neurostimulation, in which the richness of the HR model’s dynamics is matched by controllers capable of operating reliably across diverse firing regimes, network scales, and uncertainty conditions.

Appendix A

Cubics

A.1 Cubic Formula

The general form of the cubic equation is:

$$ax^3 + bx^2 + cx + d = 0 \quad (\text{A.1})$$

By applying the substitution $x = t - \frac{b}{3a}$ it is possible to obtain a depressed cubic as follows:

$$\begin{aligned} x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} &= 0 \\ \left(t - \frac{b}{3a}\right)^3 + \frac{b}{a}\left(t - \frac{b}{3a}\right)^2 + \frac{c}{a}\left(t - \frac{b}{3a}\right) + \frac{d}{a} &= 0 \\ t^3 - 3t^2\frac{b}{3a} + 3t\frac{b^2}{9a^2} - \frac{b^3}{27a^3} + \frac{b}{a}\left(t^2 - 2t\frac{b}{3a} + \frac{b^2}{9a^2}\right) + \frac{c}{a}\left(t - \frac{b}{3a}\right) + \frac{d}{a} &= 0 \\ t^3 + \frac{3ac - b^2}{3a^2}t + \frac{2b^3 - 9abc + 27a^2d}{27a^3} &= 0 \end{aligned}$$

In compact form:

$$t^3 + pt + q = 0 \quad (\text{A.2})$$

Now using the *Vieta* method, it is possible to use the substitution $t = w - \frac{p}{3w}$, from which it yields:

$$\begin{aligned} \left(w - \frac{p}{3w}\right)^3 + p\left(w - \frac{p}{3w}\right) + q &= 0 \\ w^3 - 3w^2\frac{p}{3w} + 3w\frac{p^2}{9w^2} - \frac{p^3}{27w^3} + pw - \frac{p^2}{3w} + q &= 0 \\ w^3 - \frac{p^3}{27w^3} + q &= 0 \end{aligned}$$

By multiplying both sides by w^3 it becomes a quadratic:

$$(w^3)^2 + q(w^3) - \frac{p^3}{27} = 0 \quad (\text{A.3})$$

with solutions

$$w^3 = -\frac{q}{2} \pm \sqrt{\frac{q^2}{4} - \frac{p^3}{27}} \quad (\text{A.4})$$

Thus obtaining w as:

$$w = \sqrt[3]{-\frac{q}{2} \pm \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} \quad (\text{A.5})$$

Now, the 3 solutions will be $w, we^{2i\pi/3}, we^{4i\pi/3}$, while by substituting back

$$t = w - \frac{p}{3w} \quad x = t - \frac{b}{3a} \quad (\text{A.6})$$

it holds

$$\begin{aligned} x_0 &= w - \frac{p}{3w} - \frac{b}{3a} \\ x_1 &= we^{2i\pi/3} - \frac{p}{3we^{2i\pi/3}} - \frac{b}{3a} \\ x_2 &= we^{4i\pi/3} - \frac{p}{3we^{4i\pi/3}} - \frac{b}{3a} \end{aligned} \quad (\text{A.7})$$

A.2 Discriminant

Definition A.2.1 (Discriminant). The *discriminant* of a polynomial is a function of its coefficient that is zero if and only if the polynomial has a multiple root, or, if it is divisible by the square of a non-constant polynomial.

The discriminant of a generic cubic is

$$18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2 \quad (\text{A.8})$$

To prove the preceding formulas, one can use Vieta's formulas to express everything as polynomials in r_1, r_2, r_3 , and a .

If the coefficients of a polynomial are real numbers, and its discriminant Δ there are 3 cases:

- if $\Delta > 0$ the equation has 3 distinct real roots
- if $\Delta = 0$ the equation has multiple roots (double or triple root)
- if $\Delta < 0$ the equation has one real root and 2 complex conjugates ones.

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